

BEYOND INDIVIDUAL CLAIMS: PROBABILISTIC LOGICAL COHERENCE FOR LONG-FORM LLM EVALUATION

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ABSTRACT

Long-form factuality assessment decomposes a response into atomic claims and verifies each against retrieved evidence. The paradigm is effective and now standard, but it embeds an assumption — that the truth of each claim can be settled in isolation — which discards the structure that makes prose an argument. A response can therefore pass per-claim verification while contradicting itself, and two responses built from *identical* claims can differ in coherence through the order in which they assert them. We formalize *logical coherence* as a property of the joint distribution over claim truth values induced by a Markov random field whose pairwise factors are the relations the response itself draws. The relation vocabulary is a two-level taxonomy: twelve interpretable discourse senses, grounded in PDTB 3.0 and RST, compile through a deterministic map onto five inferential couplings, each fixing a factor table by naming the joint assignment it penalizes. We characterize the tables coherence requires by a row-stochasticity property and show that those used for claim–evidence factuality are wrong here, collapsing the measure’s dynamic range; claim priors come from an existing factuality pipeline, making coherence and factuality two stages over one decomposition. On a corpus of response families whose coherence ordering is fixed by construction, the measure recovers 82.7% of declared orderings from gold relation graphs and 34.7–37.1% from graphs mined out of the response text — and since the two conditions differ only in the relation graph, that gap localizes the open problem in relation extraction rather than in the probabilistic model. Against five families of evidence-free baselines it is the only measure that both orders responses correctly (37/50 against the best baseline’s 30/50) and stays exactly invariant under meaning-preserving edits (20/20), the latter *by construction*: reordering sentences leaves the Markov network identical, a guarantee no surface statistic can offer and one that an LLM judge violates while swinging 42 points between judge models on identical prose. The score also decomposes into per-claim marginals, so an incoherence is located rather than merely detected.

1 INTRODUCTION

The dominant paradigm for assessing the factuality of long-form generation is decompose-then-verify: split a response into atomic claims, check each against retrieved evidence, and aggregate the verdicts (Min et al., 2023; Wei et al., 2024; Song et al., 2024). It is effective, it is now standard, and it embeds an assumption. Treating each claim as independently checkable discards precisely the structure that distinguishes an *argument* from a *list*.

Two failure modes make this concrete. A report may assert both that no passengers were harmed and that a wire service reported three deaths; per-claim verification reports one claim true and one false, but not the more basic defect — that the response contradicts *itself*. Separately, the inferential links a response draws may fail even when their endpoints are individually true: a chain of accurate statements joined by a spurious “therefore” is not a sound argument, and every atom in it verifies. Xie et al. (2026) quantify how often this matters: on logic-related samples, SAFE, VeriScore and FactCheck-GPT omit the logical dimension on 60%, 78% and 48% of cases respectively. The two axes are orthogonal — a response can be a bag of true facts that is incoherent, or internally impeccable while entirely false — and our sharpest fixture is a pair of summaries of one appellate judgment built from the *same* eighteen claims, one a permutation of the other, differing only in whether self-serving

assertions precede or follow the findings that qualify them: every per-atom check returns an identical verdict, yet our measure separates them (0.95 against 0.90).

We therefore ask: how much of what a response asserts does its argument, taken as a whole, uphold? That is answered by a *marginal* in a graphical model whose variables are claim truth values and whose factors are the asserted relations — not by counting violated rules, which cannot propagate a conflict from its endpoints to what they support, and which yields a measure a response maximizes by asserting no relations at all (Prop. 3).

Our contributions are as follows. (i) A **coherence MRF and a formal two-level relation taxonomy** (§3.1–§3.2): twelve discourse senses compile through a deterministic map onto five inferential couplings, each defined by the joint assignment it penalizes, which fixes its factor table — separating the interpretable layer from the inferential one lets the annotation vocabulary be rich while the model stays a closed three-parameter pairwise family. (ii) A **characterization of why factuality’s factor tables are wrong for coherence** (§3.3): they charge a premise for being a premise, and the alternative collapses the measure’s dynamic range by more than half. (iii) A **selected readout and a two-stage composition with factuality** (§3.4, §3.6): of four candidates evaluated by exact inference against five criteria, only two are monotone in conflict resolution. (iv) An **evaluation separating relation-extraction error from scoring error** (§4), by holding the readouts byte-identical and swapping only the relation graph: the measure satisfies 82.7% of 202 declared constraints from gold graphs against 34.7–37.1% from mined ones, localizing the open problem in extraction; and against five families of evidence-free baselines it is the only one that both orders responses correctly (37/50 against the best baseline’s 30/50) and is exactly invariant under meaning-preserving edits (20/20) — the latter *by construction*, a guarantee no surface statistic can offer.

2 RELATED WORK

Atomic factuality. FActScore (Min et al., 2023) introduced per-atom precision against a knowledge source; SAFE (Wei et al., 2024) added search-augmented verification and an $F_1@K$ metric, VeriScore (Song et al., 2024) restricted attention to *verifiable* claims, and FLEEK (Bayat et al., 2023) retrieves evidence to detect and correct errors. All aggregate independent per-claim verdicts and are therefore blind to inter-claim structure by construction. FactReasoner (Marinescu et al., 2025) replaced verdict counting with probabilistic inference over a factor graph of claim and evidence variables with pairwise factors from NLI relations; it is the closest antecedent — we reuse its decomposition, relation extractor and inference stack, and take its marginals as our priors (§3.6) — but its factors run between a claim and the *evidence* retrieved for it, so its graph has no claim–claim edges and its marginals cannot express internal consistency. Xie et al. (2026) states the gap directly, classifying inter-atom dependencies as a labelling task over claim pairs; our departure is to treat them as *factors in a joint distribution*, which lets a local conflict propagate to distant claims and makes the score a functional of the whole graph rather than a per-edge average.

Discourse coherence. Entity-grid models (Barzilay & Lapata, 2008) and their neural successors (Li & Jurafsky, 2017) score coherence from entity-mention distributions, evaluated against *synthetic perturbations* — a methodology we extend in §4.1 by making the perturbation an operator on a declared relation graph rather than on raw text, so its intended effect is known per item. DiscoScore (Zhao et al., 2023) uses BERT-based discourse representations. What these share is that coherence is read off surface statistics or learned representations: none exposes an inspectable relation graph, and none yields per-claim attributions. Our inventory takes its senses from PDTB 3.0 (Prasad et al., 2019) and RST (Mann & Thompson, 1988) but treats them as an interpretable layer compiling to a smaller inferential vocabulary. Closest to our mining stage is Liu & Strube (2025), who label sentence pairs with PDTB 3.0 senses via a discourse parser (App. E).

Consistency checking, judges, and relational models. SelfCheckGPT (Manakul et al., 2023) detects hallucination by pairwise NLI over sampled responses — the natural ablation isolating what global inference adds — and ROSCOE (Golovneva et al., 2023) scores reasoning chains with a self-consistency term. G-Eval (Liu et al., 2023) is the practical incumbent, an LLM judge scoring coherence from a rubric; judges are strong on aggregate correlation and weak on per-claim attribution and on any guarantee that a meaning-preserving edit cannot change the score. Steen & Markert (2022) argue coherence measures must be checked against system-level confounders, which we adopt

in our ladder-blind controls. Markov logic (Richardson & Domingos, 2006) is the natural alternative encoding; App. H shows the obvious one-clause-per-relation encoding is *not* equivalent to ours and is degenerate at its mode, while a three-clause expansion reproduces it exactly on the three-coupling fragment. We keep the pairwise MRF because its factor family is closed, and the Markov logic view because the beyond-pairwise rules addressing our main limitation live there.

3 PROBABILISTIC LOGICAL COHERENCE

This section builds the measure in six steps. We define the probabilistic object — a Markov random field over claim truth values whose factors are the relations a response draws (§3.1) — then fill in the two things that definition leaves open: *which* relations the vocabulary admits (§3.2) and *what factor table* each contributes (§3.3). With the model fixed, §3.4 reads a score off it, §3.5 estimates the relations from text, §3.6 couples coherence to factuality through the free prior slot, and §3.7 works a five-claim pair end to end. Each step closes a choice the previous one left open, and the two that matter most — the coupling vocabulary and the factor tables — are exactly where a coherence model diverges from a factuality one.

3.1 THE COHERENCE MARKOV RANDOM FIELD

Definition 1 (Coherence MRF). *Let $\mathcal{A} = \langle a_1, \dots, a_n \rangle$ be the atomic claims of a response in assertion order, each binary with $a_i = 1$ read as “the claim holds”. Let $\mathcal{R} = \{(s_r, t_r, \tau_r, p_r)\}_{r=1}^{|\mathcal{R}|}$ be a set of relations, where $s_r \neq t_r$ index claims, τ_r is an inferential coupling (Def. 2) and $p_r \in [0, 1]$ is the relation’s weight. Given priors $\pi \in [0, 1]^n$, the coherence MRF induced by $(\mathcal{A}, \mathcal{R}, \pi)$ is*

$$\mathbb{P}(\mathbf{a}) = \frac{1}{Z} \prod_{i=1}^n \phi_i(a_i) \prod_{r \in \mathcal{R}} \psi_{\tau_r, p_r}(a_{s_r}, a_{t_r}), \quad Z = \sum_{\mathbf{a} \in \{0,1\}^n} \prod_i \phi_i(a_i) \prod_r \psi_r, \quad (1)$$

where $\phi_i(a_i) = [1 - \pi_i, \pi_i]$ over the states $(0, 1)$ and $\psi_{\tau, p}$ is given by §3.3.

Three structural properties are deliberate commitments, each load-bearing later.

Remark 1 (The factor family is closed). *Every $\psi_{\tau, p}$ is determined by p , the source prior π_s and the penalized set, so the whole inferential vocabulary is a three-parameter pairwise family — a guarantee about what the model cannot do: no relation type, however elaborately named at the discourse level, can smuggle in expressiveness beyond a pairwise potential.*

Remark 2 (Order-sensitivity enters only through the relations). *Nothing in Eq. 1 references claim position; assertion order enters exclusively through which pairs are related and in which direction. An edit preserving the relation set — reordering claims whose relations are symmetric, or exchanging a Precedence for a Succession, which compiles to no factor — therefore provably cannot change any readout, a falsifiable prediction the invariance ladders of §4.1 test.*

Remark 3 (Coherence is evaluated without evidence). *The graph contains claim variables only, and with uniform priors $\pi_i \equiv \frac{1}{2}$ the model consults no external knowledge whatever: it scores how the response hangs together, not whether it is true, which is what makes the two axes separable.*

3.2 TWO LEVELS: INFERENTIAL COUPLINGS AND DISCOURSE SENSES

Definition 2 (Inferential couplings). *A coupling τ is specified by the set $V(\tau) \subseteq \{0, 1\}^2$ of joint assignments (a_s, a_t) it penalizes — the configurations at which the relation is violated:*

Coupling	Reading	Directed	$V(\tau)$
ENTAILMENT	$a_s \Rightarrow a_t$	yes	$\{(1, 0)\}$
CONTRADICTION	$a_s \Rightarrow \neg a_t$	yes	$\{(1, 1)\}$
EQUIVALENCE	$a_s \Leftrightarrow a_t$	no	$\{(0, 1), (1, 0)\}$
EXCLUSIVE	exactly one of a_s, a_t holds	no	$\{(0, 0), (1, 1)\}$
CO_NECESSITY	at least one of a_s, a_t holds	no	$\{(0, 0)\}$
NONE	no coupling	—	$\emptyset$ (no factor)

Write $\mathcal{T}$ for the five edge-producing couplings and $\mathcal{T}_{\text{conf}} = \{\text{CONTRADICTION}, \text{EXCLUSIVE}\}$ for the conflict couplings, whose penalized set contains the both-true world.

The first three are exactly the NLI edge types of an existing factuality pipeline (App. A), which lets a mined relation reuse the existing graph machinery unchanged. The last two are additions required by argumentative prose: CONTRADICTION forbids only the both-true world, whereas *exhaustive* alternatives also forbid the both-false one — “no one was harmed” and “three people died” can neither both hold nor both fail, so treating them as mere opposition leaves a world in which *neither* is true and under-penalizes — and CO_NECESSITY is its dual (App. B).

Couplings are the right vocabulary for a factor table and the wrong one for annotation, since what is observable in the text is a discourse fact rather than a logical one. We therefore mine an interpretable layer and compile it down through a total, deterministic map $\mathcal{C}$ assigning each sense a coupling, a strength prior ($\perp$ meaning “use the estimator’s own value”) and flags:

Level-2 sense	$\mathcal{C}$ coupling	prior	note
Cause-Effect*, Effect-Cause*, Evidence*	ENTAILMENT	$\perp$	directed
Condition, Instantiation	ENTAILMENT	$\perp$	directed; inadmissible (§3.5)
Restatement*	EQUIVALENCE	0.90	definitional identity
Contrast*	CONTRADICTION	$\perp$	opposed, <i>not</i> exhaustive
Concession*	CONTRADICTION	$\perp$	a tension the text may resolve
Alternative*	EXCLUSIVE	$\perp$	exhaustive: exactly one holds
Disjunction*	CO_NECESSITY	$\perp$	at least one holds
Precedence*, Succession	NONE	—	ordering-only: no factor

Five senses collapse onto ENTAILMENT, a deliberate loss: a pairwise potential over truth values cannot distinguish causation from evidence. What survives compilation is the inferential force, the sense being retained for interpretability. The two ordering-only senses draw a relation with no truth-functional content — exactly what makes an order-only edit provably score-invariant (Rem. 2).

A Concession asserts a tension and then overrides it, and treating that as a live contradiction would penalize a response for doing what careful argument requires. When a resolving *holding* claim h is present we discount the conflict’s weight,

$$p_r^{\text{eff}} = p_r (1 - \lambda \pi_h), \quad (2)$$

with $\lambda = 0.45$; absent a resolver the tension counts at full strength. Holdings are found by adjudicative cue phrases on a non-endpoint claim maximizing lexical overlap with the endpoints — a heuristic which, with λ , is the one place hand-set constants enter.

3.3 THE PAIRWISE POTENTIALS, AND WHY FACTUALITY’S TABLES ARE WRONG HERE

A coupling names the world it penalizes; it does not say by how much. Filling that in is the second half of Def. 1, and it is where a coherence model parts company with the factuality model it otherwise reuses.

Listing $\psi_{\tau,p}(a_s, a_t)$ row-major over $(0, 0), (0, 1), (1, 0), (1, 1)$, with π_s the source prior and bold marking $V(\tau)$:

ENTAILMENT	CONTRADICTION	CO_NECESSITY	EQUIVALENCE	EXCLUSIVE
$[1 - \pi_s, \pi_s, \mathbf{1 - p}, p]$	$[1 - \pi_s, \pi_s, p, \mathbf{1 - p}]$	$[\mathbf{1 - p}, \pi_s, \pi_s, p]$	$[p, \mathbf{1 - p}, \mathbf{1 - p}, p]$	$[\mathbf{1 - p}, p, p, \mathbf{1 - p}]$

Proposition 1 (Row-stochasticity). *Each table is row-stochastic in the source state: for $\tau \in \{\text{ENTAILMENT}, \text{CONTRADICTION}, \text{CO_NECESSITY}\}$ both rows sum to one, giving $\mathbb{P}(a_t=1 \mid a_s=0) = \pi_s$ and $\mathbb{P}(a_t=1 \mid a_s=1) = p$ for ENTAILMENT, CO_NECESSITY and $1 - p$ for CONTRADICTION. Consequently the factor asserts something about the target only when the source is true; when the source is false, the target returns to its prior.*

The factuality literature instead uses a *no-priors* family whose ENTAILMENT table is $[p, p, \mathbf{1 - p}, p]$: the $(0, \cdot)$ cells are both p , so the table charges the *source* being true whenever the target is uncertain. That suits claim-evidence factuality, where the source is a retrieved passage whose own reliability is in question, but it is wrong for coherence, where the source is another claim of the same response and a well-supported premise must not be penalized for being a premise. The consequence is not a matter of degree: on the sixteen-claim recall report of App. F, joined by a twelve-edge entailment backbone,

the no-priors tables drive the root to $\mathbb{P}(a_1=1) = 0.19$, far below its 0.5 prior, and the mean marginal sits at 0.490, moving only to 0.498 when every conflict edge is deleted — a range of 0.008 against 0.607 to 0.620 under the with-priors tables, all exact by 2^{16} enumeration. A coherence measure built on the factuality tables spends its dynamic range penalizing premises and has little left for conflicts.

3.4 THE COHERENCE SCORE

The model is now fully specified: Def. 1 gives the joint, §3.2 the admissible relations, §3.3 their tables. What remains is to summarize the resulting distribution in a single number — and, as it turns out, not every natural way of doing so is admissible.

We report the *mean posterior marginal* $\text{LCS}_{\text{mm}}(y) = \frac{1}{n} \sum_{i=1}^n \mathbb{P}(a_i = 1)$, read from a single MAR query: the expected fraction of the response’s claims that the argument, taken as a whole, upholds. A conflict suppresses the marginals of its endpoints, a satisfied entailment chain propagates support along itself, and a claim no relation touches stays at its prior. Three diagnostics come free with the same query: the per-claim marginals, which *localize* an incoherence; the count of claims dragged *below their own prior*; and the mean normalized binary entropy.

App. C defines three alternatives — a two-term consistency probability, a reified coherence node and a normalized log-partition — states five criteria, and evaluates all four by exact inference. Only the mean marginal and the normalized log-partition are monotone in conflict resolution, and the two failures share one cause: they measure whether a conflict is *active* rather than what should be *believed*, and softening a resolved concession by Eq. 2 necessarily *increases* the probability of its both-true world (Prop. 4). A fifth candidate — the probability that every asserted relation is satisfied — is maximized by asserting no relations whatever (Prop. 3). We select the mean marginal: it satisfies all five criteria, needs one query, and alone decomposes into per-claim contributions.

3.5 ESTIMATING RELATIONS FROM THE RESPONSE

Everything so far has taken the relation set $\mathcal{R}$ as given, which is what makes the model’s properties provable. In deployment it is not given.

In use $\mathcal{R}$ must be estimated from the text, and it is the weakest link in the pipeline (§4). A relation’s weight factors into a type term and a strength term, $p_r = \mathbb{P}(\tau \mid a_s, a_t, y) \cdot \mathbb{P}(a_t \mid a_s, \tau, y)$, estimated by two separate prompts (verbatim in App. J); conflating them is a mistake, since a confidently-identified weak link and a doubtfully-identified strong one are different epistemic situations one number cannot distinguish. *Prompt A* elicits the discourse sense with chain-of-thought *preceding* the label, so the token logprobs measure a decision already made, and the type confidence is the per-token geometric-mean probability of the label span. *Prompt B* elicits the conditional strength as a surrogate token, the renormalized $\mathbb{P}(\text{Yes}) / (\mathbb{P}(\text{Yes}) + \mathbb{P}(\text{No}))$ (Kadavath et al., 2022), asked in graded form so a weak but real link is not forced to answer No; a verbalized variant is kept only for comparison, verbalized confidence being weakly calibrated (Xiong et al., 2024).

Grounding is mandatory, not optional. Prompt A always receives the full response and asserts a coupling only where the response *itself* draws the connection — the single most consequential decision in the estimator. Judging a pair in isolation invites any abstractly plausible relation, and the failure is not mild: relation density reaches six to nine per claim, including spurious contradictions on paragraphs containing none. Since Eq. 1 is a product over relations, many soft mutually-constraining edges drive every marginal toward the prior, so an over-connected graph destroys the measure’s ability to distinguish coherent from incoherent rather than merely adding noise. Two mechanisms add precision: the *admissibility filter* restricts output to the nine senses marked $\star$ above, excluding Condition and Instantiation as broad enough that a model reaching for “related somehow” lands on them readily (13% of one arm’s edges, zero in gold); and the *evidence-span requirement* obliges the model to quote the linking words verbatim, with the parser verifying the quotation really is a substring — the only precision mechanism that is *checked* rather than asked for.

Candidate pairs. Querying all $n(n-1)$ ordered pairs is quadratic and actively harmful, so a policy restricts which are scored: ALL_PAIRS; WINDOWED, forward pairs with $0 < j - i \leq w$ ($w = 4$); GATED, the window plus similar long-range forward pairs; and BIDIRECTIONAL, both arc directions within $|j - i| \leq w$. The last exists because of a structural limitation, not a tuning preference: a

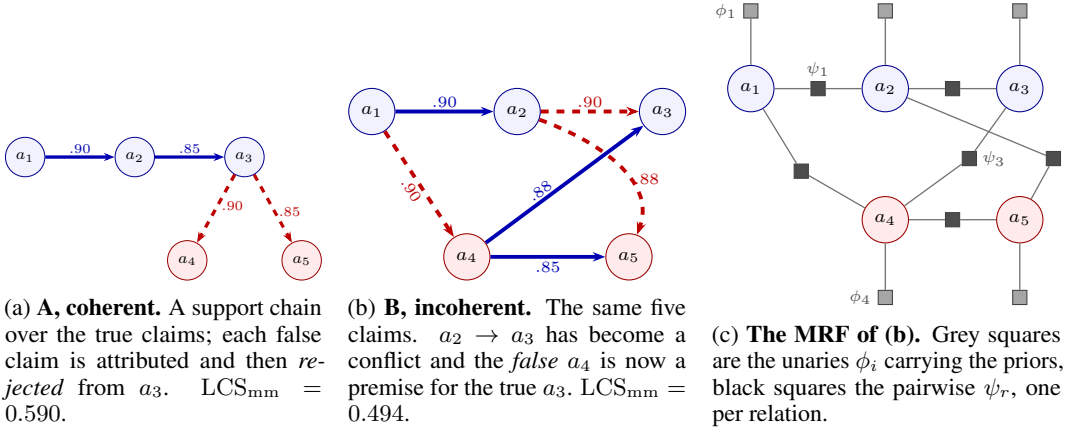


Figure 1: The five-claim example of §3.7. Blue nodes are claims true of the world ($\pi_i = 0.9$), red nodes false ($\pi_i = 0.1$); *the same five claims appear in both graphs*, so every per-claim factuality check returns the same verdict on both responses. Solid blue edges are ENTAILMENT, dashed red CONTRADICTION; line width grows with the relation weight p . The unary factors ϕ_i of panel (c) are identical for the two responses, so the entire $0.590 \rightarrow 0.494$ drop is produced by the pairwise ψ_r — by how the response arranges its claims, which is the whole of what coherence measures.

forward-only policy *cannot emit* a backward arc, so a gold relation that is both directed and backward is unreachable however the model answers — and 105 of our corpus’s 603 edge-producing gold relations are of that kind, capping forward-only recall at 82.6%. Each policy is then refined against the response, promoting discourse-adjacent out-of-window pairs and demoting in-window pairs that are not (App. D).

3.6 COMPOSING COHERENCE WITH FACTUALITY

The priors π are a free slot, and the natural thing to put in them is what a factuality pipeline already computed: $\pi_i \leftarrow q_i = \mathbb{P}(a_i = 1 \mid \mathcal{C})$ for retrieved evidence $\mathcal{C}$. A claim is then credited when it is both externally supported and internally upheld, and an unsupported claim enters the coherence graph already doubted, so an argument resting on it is penalized twice. Setting $q_i \equiv \frac{1}{2}$ recovers the pure coherence score of Rem. 3, so the composition is an option rather than a commitment. We keep two stages rather than one fused network for reasons of cost, ablation and caching that App. D sets out.

3.7 EXAMPLE

The pieces are easiest to see at work on a case small enough to enumerate exactly. This example also makes the orthogonality claim of §1 concrete, since it holds the claim set — and therefore every factuality verdict — fixed while varying only the arrangement.

Five claims about Voyager 1 are fixed once and for all, three true of the world (priors 0.9) and two false (0.1): a_1 , it crossed the heliopause in August 2012; a_2 , after the crossing its instruments measured a sharp rise in plasma density; a_3 , it is therefore in interstellar space; and, false, a_4 , its plasma wave instrument failed permanently before the crossing, and a_5 , it has left the Solar System entirely.

Two responses assert *these same five claims* and differ only in how they arrange them, so every per-claim factuality check returns the same verdict on both. **Response A** reports the density rise as what established the crossing, concludes the probe is in interstellar space, then names the two false claims and refuses each on a stated ground. **Response B** instead denies that the probe counts as being in interstellar space, asserts the instrument failure as fact, offers *that failure* as the reason the probe reached interstellar space, chains it into having left the Solar System, and finally denies both falsehoods in its closing sentence.

Figure 1 shows the resulting graphs and the MRF that B induces. Two relation changes carry the difference: the support edge $a_2 \rightarrow a_3$ in A becomes a CONTRADICTION in B, which denies its own

conclusion, and the edge between a_3 and a_4 *reverses* — A rejects the false a_4 from a_3 , while B makes a_4 a premise *for* a_3 . Mentioning a falsehood is not what costs B its score; asserting it and reasoning from it is. Exactly, by 2^5 enumeration, A scores $\text{LCS}_{\text{mm}} = 0.590$ against B’s 0.494: in A the true claims sit at 0.939, 0.989 and 0.992, each *above* its 0.9 prior because the chain supplies mutual support, and the false ones at 0.013 and 0.020, whereas in B the *true* a_3 is dragged to 0.513 because its only support arrives from a_4 , which the response itself denies — a premise the text refuses cannot transmit support. The priors enter only through the unary ϕ_i , identical in both responses, so the entire drop is attributable to arrangement alone. App. F gives both responses in full and two further examples.

4 EXPERIMENTS

The measure of §3 has two components that can fail independently: the probabilistic model, and the estimator that supplies its relation graph. An evaluation conflating them cannot say which to fix, so our experiments are built around one device — holding the readouts byte-identical and swapping only the relation graph (§4.1) — which attributes any shortfall to one component or the other by construction. We then ask three questions in turn: how accurate is the estimator (§4.2), how well does the model order responses given a correct graph (§4.3), and how does the measure compare against evidence-free alternatives (§4.4). §4.5 draws the answers together and §4.6 bounds them.

4.1 PROTOCOL

Validating a coherence measure requires responses that differ in coherence while holding content fixed. Natural corpora supply neither that nor a gold relation graph, which is what separates *scoring* error from *estimation* error. We therefore evaluate on synthesized *coherence ladders*: five responses over one claim set — a base realizing a declared relation plan, plus four variants from perturbation operators on that plan — each carrying *its own* gold graph derived by the same operators. Each family declares the ordering its rungs must exhibit: **C1** strict increase, **C2** a *predicted* decrease or invariance (where the concession dip of Prop. 4 becomes falsifiable and an ordering-only edit must be exactly flat by Rem. 2), and **C3** endpoint separation.

An *arm* fixes where the relation graph comes from while holding claims, priors, readouts and scorer settings byte-identical — the central experimental device of the paper. **GOLD** takes each item’s own labelled relations, **GOLD_VALID** only those labelled valid, and a **MINED** arm runs the estimator of §3.5 on the response text; because the readouts are identical across arms, any shortfall below gold is attributable to relation estimation and not to the probabilistic model. Mined relations are compared at **pair, coupling** (plus direction, where asymmetric) and **sense** level, coupling being where a factor table is selected.

The corpus holds **fourteen families and seventy items**, produced by **claude-opus-5** and admitted by a three-model cross-vendor committee (**gpt-5.6-terra**, **gemini-3.1-pro**, **claude-sonnet-5**) that recovers the graph from prose without seeing the plan, audits for leakage and checks atom coverage. Items carry 15–16 claims and 609–698 words, for 1,115 atoms and 670 gold relations at a 0.61 valid fraction, plus 342 pairs the corpus *explicitly declares unrelated*. Eight families are **CONFLICT**, two **CHAIN**, three **ORDER** and one **CONTROL**; the last two apply only meaning-preserving edits and so assert *invariance*, without which a corpus cannot distinguish a coherence measure from an edit-count proxy. The declarations total 202 **constraint checks** (112/30/45/15). Mining is sampled once per cell: two *independent* runs moved every F1 by at most 0.014 while gold arms reproduced bit-for-bit, and ladder agreement moved by four to six of 202 assertions. Exact values are by 2^{16} enumeration, corpus-scale ones by weighted mini-bucket (Liu & Ihler, 2011) at i-bound 6.

Five baseline families are chosen to fail in informative ways; four are implemented and share the LCS’s own decomposition, so each sees the same atoms and no retrieved evidence: ladder-blind controls (claim count, relation count, length) that catch a measure secretly tracking edit count or graph density (Steen & Markert, 2022); pairwise NLI with no global model (Manakul et al., 2023); ROSCOE’s Self-Consistency (Golovneva et al., 2023) in max, mean and symmetric variants; LLM judges — G-Eval (Liu et al., 2023) with the SummEval rubric (Fabbri et al., 2021) and a direct rating

Table 1: Mined-versus-gold agreement over the 603 edge-producing gold relations (670 total, less the ordering-only edges, which contribute no factor), micro-averaged over seventy items. **Fwd/Bwd** split coupling recall by whether the gold relation runs forward or backward; 105 of the 603 run backward. **Non-rel** counts violations of the 342 declared-unrelated pairs.

Miner	Policy	Edges	Pair R	Coup P	Coup R	Coup F1	Fwd	Bwd	Non-rel
llama-3.3-70b	WINDOWED	1106	82.3	34.1	62.5	44.1	71.1	21.9	34/342
llama-3.3-70b	BIDIRECTIONAL	3172	91.7	17.9	79.3	29.3	79.9	76.2	119/342
gpt-oss-120b	WINDOWED	1094	83.4	40.2	73.0	51.9	82.1	29.5	38/342
gpt-oss-120b	BIDIRECTIONAL	2973	96.5	22.4	88.4	35.8	89.0	85.7	147/342

Table 2: Declared ordering constraints satisfied by ladder type — 202 assertions per arm. ORDER and CONTROL assert *invariance*. GOLD_VALID is near-flat on CONFLICT by construction, so its failures carry no information about the readouts; the GOLD arm is what the constraints test.

Arm	Policy	Constraints satisfied				
		Cfl	Chn	Ord	Ctl	%
GOLD	—	89/112	18/30	45/45	15/15	82.7
GOLD_VALID	—	37/112	12/30	45/45	15/15	54.0
<i>mined by llama-3.3-70b</i>						
	WINDOWED	53/112	17/30	0/45	0/15	34.7
	BIDIRECTIONAL	57/112	18/30	0/45	0/15	37.1
<i>mined by gpt-oss-120b</i>						
	WINDOWED	59/112	12/30	0/45	0/15	35.1
	BIDIRECTIONAL	53/112	17/30	0/45	0/15	34.7

prompt, each a mean over five runs; and the three single-document DiscoScore metrics (Zhao et al., 2023) with an entity grid (Barzilay & Lapata, 2008).

4.2 RELATION ESTIMATION IS THE BOTTLENECK

We begin with the estimator, since every mined arm inherits its errors. Measuring it against gold separates a model that scores a graph badly from an extractor that supplies the wrong graph.

Three findings stand out from Table 1. First, the forward-only policy is *structurally* capped: WINDOWED recovers 21.9–29.5% of backward-running gold relations against 76.2–85.7% for BIDIRECTIONAL, because 105 of the 603 scorable gold relations are both directed and backward and no forward-only candidate set can emit them. Second, admitting both directions buys that recall at a precision cost severe enough to invert the F1 ordering: it roughly triples the edges (1,106 $\rightarrow$ 3,172 for llama), lifts coupling recall by 16.8 points and *halves* precision (34.1 $\rightarrow$ 17.9), so coupling F1 falls from 44.1 to 29.3. Because Eq. 1 is a product over relations, spurious factors drive marginals toward the prior rather than merely diluting a signal; the 1,231 duplicate unordered pairs are the same defect seen structurally — two factors on one variable pair, which the model has no principled way to combine — and declared-non-relation violations rise from 34/342 to 119/342 in step. Third, gpt-oss-120b is stronger at every level — by 7.8 points of coupling F1, 9.3 of sense F1 and 10.5 of backward recall on WINDOWED, while emitting slightly *fewer* edges — though its strength estimates are 99% saturated at a numerical boundary against 58% interior for llama, so it supplies almost no graded uncertainty.

4.3 THE MODEL ORDERS CORRECTLY WHEN THE GRAPH IS CORRECT

Holding the readouts fixed and varying only the graph now isolates the model’s own behaviour.

On gold relations the readouts satisfy 82.7% of the declared constraints — 167 of 202 (Table 2) — against 34.7–37.1% for mined arms. Since the readouts, claims and priors are byte-identical across arms, the 46–48 point gap is attributable to relation estimation alone, and it is consistent with Table 1: at 29–52% coupling F1 roughly half the factors in a mined network are wrong or missing, and a

Table 3: Both axes at once. **Increase** is the 50 readout-independent ordering assertions over the eight CONFLICT and two CHAIN families; **Invariance** the 20 rung pairs over the three ORDER and one CONTROL families, satisfied when the score is unchanged to 10^{-6} . The three baselines at 20/20 invariance reach it *vacuously*, being constant over all seventy items, which is why the final column takes the weaker axis.

Measure	Increase (of 50)		Invariance (of 20)		min
LCS _{mm} (gold relations)	37	74	20	100	74
LCS _{lp} (gold relations)	36	72	20	100	72
LCS _{cn} (gold relations)	34	68	20	100	68
RC (cohesion)	30	60	16	80	60
LC (cohesion)	28	56	16	80	56
LENGTH (control)	20	40	12	60	40
ROSCOE-SC (mean)	16–17	32–34	5–9	25–45	32
CONTRADICTION RATE	12	24	8–9	40–45	24
JUDGE, G-Eval	0–21	0–42	5–13	25–65	0
JUDGE, direct rating	4–18	8–36	5–14	25–70	8
ROSCOE-SC (max)	0	0	20	100 [†]	0
ENTITY GRAPH	0	0	20	100 [†]	0
CLAIM COUNT (control)	0	0	20	100 [†]	0

[†] vacuous: the measure is constant across all seventy items.

ladder whose consecutive rungs differ by one or two relations cannot survive that. This is the paper’s central empirical claim, and it is a claim about *where the remaining problem is*: the probabilistic model orders responses correctly when handed a correct graph, and the open problem is extracting one.

The per-ladder breakdown localizes the failure. On gold, ORDER and CONTROL are perfect (45/45, 15/15) since those rungs move no factor, CONFLICT is strong at 89/112, and CHAIN the weak ladder at 18/30. But *every mined arm scores 0/45 on ORDER and 0/15 on CONTROL* — the most consequential number in the table, and not a scoring artefact: those ladders hold the claim set fixed and permute the prose, yet on one ORDER family the `gpt-oss` WINDOWED miner emitted 9, 16, 20, 20 and 17 edges across five rungs differing *only* in sentence order. LLM relation extraction is therefore not reorder-invariant, and any measure built on it inherits that sensitivity however sound its probabilistic layer. Mined arms are nonetheless *not* uniformly worse (CONFLICT 53–59 of 112, CHAIN 12–18 of 30), so a measure evaluated only on increase-type ladders would have called mined extraction far closer to gold than it is — precisely the false positive the CONTROL ladder was added to catch. Mined ladder agreement also does *not* track coupling F1 — `llama`’s BIDIRECTIONAL arm has the worst coupling F1 (29.3) and the best agreement (37.1%) — so these arms are not separated by ladder agreement at all, and candidate policies should not be ranked on it.

4.4 COMPARISON WITH EVIDENCE-FREE BASELINES

The gold arm establishes what the model does with a correct graph; it does not establish that the exercise needs a probabilistic model at all. That requires comparison against measures which see the same atoms and no retrieved evidence.

Only the readout-independent content of a constraint is scorable by a single-score baseline, leaving 50 increase assertions and 20 invariance pairs; a measure is useful only if it passes both.

Given a correct relation graph the LCS is the only measure here that passes both axes non-vacuously: 37/50 on ordering and 20/20 on invariance, with all three graded readouts above every evidence-free baseline.

Every model-free baseline is bit-identical across the two judge runs — the internal check that the harness is deterministic where it should be — and against that fixed background the judge rows move enormously: G-Eval 0/50 against 21/50, direct rating 4/50 against 18/50, on identical prose and rubric.

4.5 SUMMARY: WHAT THE MEASURE BUYS OVER THE BASELINES

Three results combine into one claim: the probabilistic layer is sound (82.7% of 202 constraints on gold graphs, 45/45 and 15/15 where exact flatness is required); it is not the bottleneck (mined graphs fall to 34.7–37.1%); and the measure already beats every evidence-free alternative we ran, all three graded readouts sitting above the best baseline’s 60%. The baselines moreover fail for three different reasons, which is why the gap is one of kind rather than degree. *Cohesion metrics* are the strongest and most instructive: they reach 60% and 56%, yet cannot separate a response that contradicts itself from one that does not when both repeat the same nouns — a neighbouring construct, not this one measured badly. *Contradiction-based metrics* lose conflict in aggregation, tracking response size; *LLM judges* move 0/50 $\rightarrow$ 21/50 on a change of judge alone; and the three baselines at 20/20 invariance are constant across all seventy items. Two properties of the model explain the gap, neither a matter of tuning. Invariance is *structural* — permuting sentences changes neither which claims stand in which relation nor the *Precedence* $\leftrightarrow$ *Succession* compilation to NONE, so the Markov network is *literally identical*, whereas every baseline here is a function of the surface. And the score *decomposes*: as a mean of per-claim marginals, the count of claims dragged below their own prior localizes an incoherence to the claims responsible — which no rubric scalar can do.

4.6 LIMITATIONS

Three of the results above are strong enough to be worth bounding precisely. The invariance guarantee holds only given a correct graph, the pairwise factorization excludes a class of real incoherence, and two constants in the design are set by hand rather than derived.

Proposition 2 (Jointly inconsistent, pairwise consistent). *Let $\mathbb{P}(\mathbf{a})$ factorize as in Eq. 1 over a_1, a_2, a_3 with non-negative factors. If $\mathbb{P}(a_1=1, a_2=1, a_3=1) = 0$ then $\mathbb{P}(a_i=1, a_j=1) = 0$ for some pair $i \neq j$, so no member of the family represents “the three claims cannot all hold, but any two can” (proof in App. G).*

Genuinely k -ary conflicts are therefore unrepresentable rather than merely approximated badly, and this is not hypothetical: a planted arithmetic inconsistency — 356 teams forming 32 conferences of exactly 10 members, where no two claims conflict but $32 \times 10 \neq 356$ requires all three — yields zero mined conflict edges and is reported as coherent, unchanged under ALL_PAIRS. Transitivity is likewise not enforced. Both defects are expressible as first-order rules in the Markov logic view of App. H, so Rem. 1’s closure guarantee is exactly what buys tractability and what costs us this defect. Two constants remain hand-set ($\lambda = 0.45$ and the resolver heuristic), and production inference is weighted mini-bucket at i-bound 6 against 2^n enumeration ($n \leq 20$), an approximation we have not characterized. At 52% coupling F1 at best roughly half a mined network’s factors are wrong or missing, so mined-arm numbers measure current extraction quality rather than the measure’s ceiling; human scoring of the invariance rungs, human correlation generally, and judge dependence (42 points between judges) remain open, and the corpus is LLM-generated throughout.

5 CONCLUSION

Coherence is a property of a joint distribution over what a response asserts, readable off a Markov random field whose factors are the relations the response itself draws. Three things follow. First, the factor tables matter: those right for grounding a claim against evidence charge a premise for being a premise, so a coherence measure built on them spends its dynamic range in the wrong place (Prop. 1). Second, the relation vocabulary can be rich at the annotation layer and must be narrow at the inferential one: twelve senses compiling onto five couplings keeps the model a closed pairwise family while leaving the labels interpretable. Third, not every reading of “how coherent is this?” is admissible: two of four candidates are non-monotone in conflict resolution because they measure whether a conflict is active rather than what should be believed, and a fifth is maximized by asserting no relations at all. Empirically the probabilistic layer is adequate and grounded extraction is not, which is where the open problem lies.

AI USE STATEMENT

Generative AI is both an object of study and a component of the artifacts released here, and we disclose both roles. The relation estimator of §3.5 is LLM-based by design: its prompts, its logprob-derived confidences and its failure modes are contributions of the paper. The evaluation corpus of §4.1 is LLM-generated throughout, including its gold relation labels, and validated by an independent LLM committee; §4.6 states the consequences for treating it as ground truth. We additionally used AI assistants for code implementation and for editorial assistance on this manuscript. All LLM-generated code was reviewed by the authors and is covered by a test suite that pins the exact numerical values reported here; every number in §4 was traced to a stored experiment record. The research questions, the taxonomy, the readout definitions, the propositions and their proofs are the authors’ own. We take responsibility for the final content of this work.

ETHICS STATEMENT

This work involves no human subjects and no personally identifying data. The fixtures include a summary of a published appellate judgment, used as a worked example of ordering-dependent coherence; it is a matter of public record and no private individual is the subject of analysis. The principal ethical risk is misreading: a coherence score is not a truth score, and a response can be perfectly coherent and entirely false. That is precisely why §3.6 keeps the two axes separable, and we caution against deploying LCS alone as a factuality filter — a fluent, internally consistent fabrication is exactly the artifact it scores highest. The released fixtures and corpus deliberately contain false claims, labelled as such, and should not be redistributed without those labels.

REPRODUCIBILITY STATEMENT

The model is fully specified in the paper: Def. 1 gives the joint, §3.3 the potentials, Def. 2 the penalized sets, §3.2 the compile map, and Eq. 4 the selected readout, with the three alternatives and the auxiliary-variable constructions two of them require in App. C. Algorithms 2 and 1 give candidate selection and the end-to-end procedure. App. J reproduces all four estimation prompts verbatim, including the trailing token boundary the surrogate readout depends on. The exact values in §3.7 and App. F are computed by explicit 2^n enumeration — 2^5 for the five-claim pair, 2^{16} for the recall report — and are pinned by regression tests in the released code, so they are reproducible without LLM access. The k -ary representability figures of App. G are likewise exact by 2^3 enumeration, with the seeded random search and the published table both reproduced by a released script. §4 names every model used and states the sampling regime and its measured variance. Code, fixtures and corpus are released.

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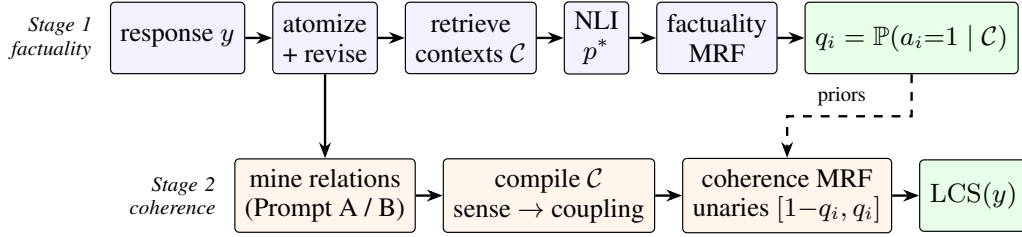


Figure 2: The two-stage pipeline. Stage 1 (§A) is FactReasoner (Marinescu et al., 2025): it grounds each atom against retrieved evidence and returns a posterior marginal q_i . Stage 2 (§3) builds a second MRF over the atoms *alone*, takes q_i as its unary priors, and adds one pairwise factor per mined atom–atom relation. Atomization is shared, so the response is decomposed once. Setting $q_i \equiv \frac{1}{2}$ recovers a pure coherence score that consults no external evidence.

A STAGE ONE: PROBABILITIES FOR INDIVIDUAL CLAIMS

The coherence model consumes claim-level probabilities and reuses the inference machinery that produces them. That stage is FactReasoner (Marinescu et al., 2025); the account here is a summary sufficient to make the composition of §3.6 precise, and it also fixes the label-span estimator whose construction the relation miner of §3.5 reuses.

A.1 DECOMPOSITION, RETRIEVAL, AND RELATION EXTRACTION

A response y is decomposed by an LLM *atomizer* into atomic claims $\mathcal{A} = \{a_1, \dots, a_n\}$, each a single independently verifiable proposition. A *reviser* then decontextualizes each one, resolving pronouns and anaphora so that the claim is checkable outside its original sentence; near-duplicates are collapsed. For each claim a retriever returns supporting passages, which are deduplicated into an evidence set $\mathcal{C} = \{c_1, \dots, c_m\}$.

An LLM-based NLI extractor then labels admitted pairs. For a claim–evidence pair the label is drawn from $\{\text{entailment}, \text{contradiction}, \text{neutral}\}$; for an evidence–evidence pair both directions are scored and reconciled, with entailment in both directions relabelled *equivalence*. Neutral verdicts are dropped rather than encoded: a claim about which the evidence is silent contributes no factor and is left at its prior, which is what makes the factor set of §3 contain only the relations someone actually asserted.

A.2 READING A PROBABILITY OFF THE GENERATION

Each retained relation carries a probability p^* . Rather than asking the model to verbalize a confidence, we read it from the distribution over the tokens it actually emitted. The extractor is prompted to end with a bracketed label; let T be the set of generated tokens overlapping the span of that label in the decoded text. Then

$$p^* = \exp\left(\frac{1}{|T|} \sum_{t \in T} \log p_t\right), \quad (3)$$

the per-token geometric-mean probability of the label the model produced. Three details of this construction are load-bearing, and we record them because each was a source of silent error.

Span alignment. The probability is measured over exactly the span the label *parser* reports, reconstructing the decoded string from the token list and tracking each token’s character offsets. Measuring the same span the label came from makes the label and its probability structurally unable to disagree. Requiring standalone bracket-delimiter tokens instead does not work: tokenizers routinely fuse `[`, `]` and `"` into subwords with adjacent characters, so a delimiter-based rule silently fails on some backends and not others.

Token-array conventions. An earlier version of this code stripped the final entry of the logprob array as an end-of-sequence token. OpenAI-compatible and vLLM logprob arrays contain only emitted content tokens — the stop is signalled out of band — so dropping the last entry deleted a real content token, and for a bracketed label that token is the one closing the label whose confidence is being measured.

Degenerate cases. When the logprob array is empty, no label span is found, or no token aligns to it, the estimator returns 0.5 rather than 0.0. Returning zero would be read downstream as a *degenerate*, *impossible* relation for a label the model in fact generated, which is a stronger claim than “we could not measure the confidence”.

Backends without logprobs. Some serving stacks expose no token logprobs at all. There p^* instead comes from SIMBA-UQ (Bhattacharjya et al., 2025), which samples the extractor repeatedly across a temperature schedule, builds a pairwise similarity matrix over the samples, and aggregates each sample’s similarity row into a self-consistency confidence. The winning sample supplies the label and its aggregated confidence supplies p^* , so the downstream model is unchanged.

A.3 THE FACTUALITY NETWORK AND ITS VARIANTS

The labelled graph becomes a pairwise MRF with one binary variable per claim and per evidence passage. Unary factors are $\phi_X(x) = [1 - \pi_X, \pi_X]$ with π_X the prior on X ; each retained relation contributes one pairwise factor from Table 7 below, oriented so the evidence passage is the premise. Marginals $\mathbb{P}(A_i=1)$ are computed by weighted mini-bucket elimination (Liu & Ihler, 2011) with i-bound 6, and a claim is reported supported when $\mathbb{P}(A_i=1) > \frac{1}{2}$.

Three graph variants differ in which factors are admitted: FR1 connects each claim only to the passages retrieved for it; FR2 deduplicates passages and scores every claim against all of them; FR3 adds evidence–evidence factors. We use FR2. Because scoring every claim against every passage is $O(nm)$ LLM calls, a candidate-pair policy can restrict the pairs actually scored; the calibration of that policy is instructive for the analogous choice in §3.5. Replaying policies against recorded verdicts on a 20-claim narrative with 60 passages, a provenance-restricted policy with an embedding similarity gate holds recall at 1.000 through a threshold of 0.20, slipping to 0.985 at 0.30 and 0.833 at 0.40. Substituting token-Jaccard similarity for sentence embeddings, however, loses 22 of 72 non-neutral relations and moves the factuality score by 0.05 — and *no* threshold recovers full recall, even 0.05. Cheap surrogates for semantic similarity are not safe here, and the saving is workload-dependent rather than constant: roughly $5\times$ on unrelated subtopics but $1.2\times$ on a narrative where every claim shares characters.

Conventions. We use the with-priors factor tables (Table 7) throughout, with $\pi_C = 0.9$ for evidence and $\pi_A = 0.5$ for claims. The published factuality model uses the no-priors tables and a higher evidence prior. The difference is immaterial to the coherence results below, which set claim priors explicitly, but the convention must be stated because the two table families are not interchangeable — that is the subject of §3.3.

B THE RELATION TAXONOMY IN FULL

§3.2 states the two-level taxonomy compactly. This appendix grounds every entry in a realistic response excerpt, gives the compile map with its glosses and flags, and works one coupling through by hand.

B.1 ONE EXCERPT PER INFERENTIAL COUPLING

Table 4 gives a response excerpt for each coupling, set against the world that coupling denies; since a coupling *is* its penalized set, reading the two together is the quickest way to see what each one commits the response to. The first three couplings are exactly the NLI edge types of §A, which is what lets a mined relation reuse the existing graph machinery unchanged. The last two are additions required by argumentative prose, and the distinction between them and CONTRADICTION is the one the taxonomy exists to draw.

Why exhaustiveness needs its own coupling. CONTRADICTION forbids only the both-true world: two claims that cannot both hold, but might both fail. EXCLUSIVE additionally forbids the both-false world, which is what *exhaustive* alternatives assert. “No one was harmed” and “three people died” can neither both hold nor both fail — either someone was harmed or no one was — so a model that treats them as mere opposition has mis-described the situation in a way that matters for scoring: it leaves

Table 4: One realistic response excerpt per Level-1 coupling of Def. 2. As in Table 6, each excerpt is prose containing both claims — A is the source, underlined, B **the target**, **bold**, the *italic* words drawing the link — and $V(\tau)$ is repeated from Def. 2 so the excerpt can be read against the world it rules out. The *Why* column names the world the response denies, which is what distinguishes each coupling from the one above it; the three conflict-adjacent couplings CONTRADICTION, EXCLUSIVE and CO_NECESSITY are illustrated on one two-part weld failure so that only exhaustiveness varies. NONE produces no factor, so its excerpt is one where the text relates the claims without constraining their truth.

Coupling	Reading	$V(\tau)$	Response excerpt, and the world it denies
ENTAILMENT	$a_s \Rightarrow a_t$	$\{(1, 0)\}$	“The alloy is chemically identical to the certified <u>reference</u>, <i>and therefore it meets the airframe specification</i>.” <i>Why</i>: the response rules out the world where the source holds and the target fails — $(1, 0)$ — and says nothing about the case where the alloy differs, which is why the $a_s=0$ row returns the target to its prior.
CONTRADICTION	$a_s \Rightarrow \neg a_t$	$\{(1, 1)\}$	“The weld <u>passed the tensile test</u>, <i>yet the same joint was porous along its full length</i>.” <i>Why</i>: only the both-true world $(1, 1)$ is denied. Both can still fail together — if the wrong joint was tested, neither claim holds — so $(0, 0)$ must stay available.
EXCLUSIVE	exactly one holds	$\{(0, 0), (1, 1)\}$	“The joint was destructively examined, so <i>exactly one of the weld met the tensile minimum and the weld fell short of it must be the finding</i> — <i>there is no third outcome</i>.” <i>Why</i>: both same-value worlds are denied. The added $(0, 0)$ is the whole difference from CONTRADICTION: once the joint has actually been examined the pair cannot both fail, so leaving that world open — as CONTRADICTION does — would under-penalize the conflict.
CO_NECESSITY	at least one holds	$\{(0, 0)\}$	“The fracture began at the joint, so <i>at least one of the weld and the fastener must have given way</i> — conceivably both.” <i>Why</i>: the dual of EXCLUSIVE. Only the both-false world is denied, so asserting the disjunction is evidence for both endpoints (Example 1).
EQUIVALENCE	$a_s \Leftrightarrow a_t$	$\{(0, 1), (1, 0)\}$	“The shipment was certified to specification — <i>that is to say, every contractual requirement had been met</i>.” <i>Why</i>: the two disagreeing worlds are denied, in both directions; unlike ENTAILMENT there is no vacuous row, since a false source now also forces a false target.
NONE	no coupling	$\emptyset$	“The replacement units were installed in March, <i>some three months before the first in-flight failures were reported</i>.” <i>Why</i>: no world is denied. The response genuinely relates the claims, but ordering them in time constrains neither one’s truth, so the pair contributes no factor and both claims stay at their priors.

available a world in which *neither* claim is true, and so under-penalizes. Conversely CO_NECESSITY is the dual, penalizing only the both-false world, and is what a disjunction asserts (“at least one of these two findings must hold”). Adjudicated disputes — incident investigations, litigation, competing hypotheses — are precisely the domains where exhaustive pairs occur, because the function of a finding is to close one. The CONTRADICTION, EXCLUSIVE and CO_NECESSITY rows of Table 4

Table 5: The compile map $\mathcal{C}$ of §3.2, with glosses. *Ordering-only* senses record assertion order but induce **no factor**, which is what makes an order-only edit provably score-invariant (Rem. 2). **Concession** is a conflict the text may itself resolve, handled by Eq. 2. The nine senses marked $\star$ form the admissible vocabulary of §3.5.

Level-2 sense	$\mathcal{C}$ coupling	prior	dir.	flag	gloss
Cause-Effect $\star$	ENTAILMENT	$\perp$	yes		source brings target about
Effect-Cause $\star$	ENTAILMENT	$\perp$	yes		source is the effect, target the cause
Evidence $\star$	ENTAILMENT	$\perp$	yes		source is evidence for target
Condition	ENTAILMENT	$\perp$	yes		source conditions target
Instantiation	ENTAILMENT	$\perp$	yes		target instantiates source
Restatement $\star$	EQUIVALENCE	0.90	no		definitional identity
Contrast $\star$	CONTRADICTION	$\perp$	no		opposed, <i>not</i> exhaustive
Concession $\star$	CONTRADICTION	$\perp$	yes	conc.	a tension the text resolves
Alternative $\star$	EXCLUSIVE	$\perp$	no		exhaustive: exactly one holds
Disjunction $\star$	CO_NECESSITY	$\perp$	no		at least one holds
Precedence $\star$	NONE	—	yes	ord.	temporal order only
Succession	NONE	—	yes	ord.	temporal order only

state one two-part weld failure three times over, so that the only thing varying across them is which same-value worlds the response leaves standing.

Example 1 (A CO_NECESSITY factor in isolation). *Take a single CO_NECESSITY relation with $p = 0.9$ over two claims with priors $\pi = 0.5$, and no other factors. From Table 7 the pairwise table is $[1 - p, \pi_s, \pi_s, p] = [0.1, 0.5, 0.5, 0.9]$, and multiplying in the unaries and normalizing gives world masses*

$$\mathbb{P}(0, 0) = 0.05, \quad \mathbb{P}(0, 1) = 0.25, \quad \mathbb{P}(1, 0) = 0.25, \quad \mathbb{P}(1, 1) = 0.45.$$

The both-false world is suppressed from the 0.25 it would receive under independent fair priors to 0.05, while the three worlds satisfying “at least one holds” share the recovered mass in proportion to how many endpoints they assert. Each marginal rises from 0.5 to 0.7: asserting that at least one of two claims holds is evidence for both, which is the intended reading.

B.2 THE COMPILE MAP, WITH GLOSSES

Couplings are the right vocabulary for a factor table and the wrong one for annotation. Asked whether two claims stand in ENTAILMENT, an annotator — human or model — has been handed a question about logic, when the fact actually observable in the text is a question about discourse: does this sentence give a cause, offer evidence, concede a tension, or lay out alternatives? We therefore mine an interpretable layer and compile it down.

B.3 ONE EXCERPT PER DISCOURSE SENSE

Table 6 grounds each sense in a response excerpt that contains both claims and the wording that licenses the link, since what an annotator must actually recognize is a form of words rather than a truth-functional pattern. Several entries encode judgements worth defending. **Restatement** carries a fixed strength prior of 0.90 because a marked paraphrase (“in other words”, “equivalently”) is a claim about identity of content, and a model asked to estimate its strength has little to add; the estimator’s value is used when available, and the prior seeds it otherwise. Five senses collapse onto ENTAILMENT, which is a deliberate loss: the model does not distinguish causation from evidence, because a pairwise potential over truth values cannot. The grouped ENTAILMENT block of Table 6 shows exactly what compilation discards — five distinct discourse facts, one factor table. What survives compilation is the inferential force, and the sense is retained alongside it for interpretability and for error analysis.

The Contrast/Alternative boundary is exactly the exhaustiveness question of §3.2, and the Contrast, Alternative and Disjunction rows of Table 6 are that question in concrete form: one two-part weld failure written up three ways — as mere opposition, as an exhaustive choice, and as a disjunction — differing in nothing but which same-value worlds the response leaves available. The two ordering-only senses are the taxonomy’s way of representing a relation the text genuinely draws while asserting

Table 6: One realistic response excerpt per Level-2 sense of Table 5, in the same row order and grouped by the coupling each compiles to. Each excerpt is prose of the kind the estimator actually sees and *contains both claims*: *A* is underlined, *B* is **bold**, and the *italic* words are the ones that draw the link — the span §3.5 obliges the model to quote, which is why it must be findable in the response and not in either claim. Senses are marked * when admissible; the three unmarked ones are included because their breadth is the argument for excluding them (§3.5). The Contrast, Alternative and Disjunction rows describe one two-part weld failure three ways, separated only by exhaustiveness (§3.2); they reuse the scenario of Table 4 deliberately, so that a discourse sense and the coupling it compiles to can be compared on the same prose.

Level-2 sense	<i>C</i>	Response excerpt, and why this sense rather than a neighbour
Cause-Effect*	ENTAILMENT	“<u>The company shipped a batch it knew to be out of tolerance</u>, and <i>as a direct result</i> its share price fell 15% that quarter.” Why: the response itself presents <i>A</i> as bringing <i>B</i> about; the marker names the direction.
Effect-Cause*	ENTAILMENT	“<u>The entire fleet was grounded on 14 March</u>. The order came <i>because</i> inspectors had found a cracked wing spar on two airframes.” Why: the same support runs from the stated effect to its cause. Direction of <i>explanation</i> is reversed; direction of inferential support is not.
Evidence*	ENTAILMENT	“<u>The maintenance log records no inspection between January and June</u>, <i>which shows that</i> the mandatory 500-hour check was skipped.” Why: <i>A</i> is offered in support of <i>B</i> without any claim that it caused it — a record is evidence, not a mechanism.
Condition	ENTAILMENT	“A part is cleared for airframe use <i>only if</i> <u>its alloy meets the reference specification</u>.” Why: <i>B</i> is asserted contingent on <i>A</i>. Note how weakly the cue constrains: nearly any dependence can be reread as a condition, which is why the sense is inadmissible.
Instantiation	ENTAILMENT	“<u>Several suppliers failed the 2023 audit</u> — <i>for instance</i>, the Tucson plant failed on two counts.” Why: <i>B</i> is one case falling under <i>A</i>’s generalization. Like Condition, broad enough to fit almost any pair that shares a topic.
Restatement*	EQUIVALENCE	“<u>The shipment was certified to specification</u> — <i>in other words</i>, every requirement in the contract had been met.” Why: a <i>marked</i> paraphrase asserts identity of content, which is why the strength prior is fixed at 0.90 rather than estimated.
Contrast*	CONTRADICTION	“<u>The weld passed the tensile test</u>, <i>but</i> the same joint was porous along its full length.” Why: opposed and <i>not</i> exhaustive. A third possibility — the wrong joint was tested — leaves both false, so the both-false world must stay available.
Concession*	CONTRADICTION	“<i>Although</i> <u>the airline argued throughout that pilot error was to blame</u>, the final report identified a metallurgical defect; the tribunal <i>ultimately held</i> for the latter.” Why: a tension the text itself settles. The resolving holding triggers the discount of Eq. 2 rather than counting as a live conflict.
Alternative*	EXCLUSIVE	“The two accounts cannot be reconciled: <i>either</i> <u>no one aboard was harmed</u>, <i>or</i> three passengers died, and <i>exactly one of them is true</i>.” Why: exhaustive. Unlike Contrast the pair can neither both hold nor both fail, so the both-false world is penalized too.
Disjunction*	CO_NECESSITY	“Metallurgy confirms the fracture began at the joint, so <i>at least one of</i> <u>the weld</u> <i>and</i> the fastener must have failed — possibly both.” Why: at least one holds and both may. The dual of Alternative: only the both-false world is penalized.
Precedence*	NONE	“The replacement units were installed in March, <i>some three months before</i> the first in-flight failures were reported in June.” Why: the text genuinely relates the two claims, and says nothing about either one’s truth — so the sense induces no factor.
Succession	NONE	“The first in-flight failures were reported in June, <i>fully three months after</i> the replacement units had been installed.” Why: the mirror of Precedence — the same two claims, the same absence of truth-functional content. Offering both senses invites an arbitrary choice on a relation that produces no factor either way.

Table 7: Pairwise potentials $\psi_{\tau,p}(a_s, a_t)$, listed row-major over the assignments $(0, 0), (0, 1), (1, 0), (1, 1)$; π_s is the source claim’s prior and bold marks the penalized set $V(\tau)$ of Def. 2. The *no-priors* family is correct for claim–evidence factuality, where the source passage’s own truth is at stake; the *with-priors* family is what coherence requires (Prop. 1) and what Def. 1 uses. The two symmetric couplings EQUIVALENCE and EXCLUSIVE coincide in both families, having no source-prior term to correct.

Coupling τ	No priors	With priors	Symmetry
ENTAILMENT	$[p, p, \mathbf{1-p}, p]$	$[1-\pi_s, \pi_s, \mathbf{1-p}, p]$	asymmetric
CONTRADICTION	$[p, p, p, \mathbf{1-p}]$	$[1-\pi_s, \pi_s, p, \mathbf{1-p}]$	asymmetric
EQUIVALENCE	$[p, \mathbf{1-p}, \mathbf{1-p}, p]$		symmetric
EXCLUSIVE	$[\mathbf{1-p}, p, p, \mathbf{1-p}]$		symmetric
CO_NECESSITY	$[\mathbf{1-p}, p, p, p]$	$[\mathbf{1-p}, \pi_s, \pi_s, p]$	symmetric

that it has no truth-functional content: a narrative that says one event preceded another has related the two claims, and has said nothing about whether either is true.

C POTENTIALS AND THE FOUR COHERENCE READOUTS

C.1 BOTH FAMILIES OF PAIRWISE POTENTIALS

§3.3 states Prop. 1 and the collapse it prevents. The table above gives both families side by side: the *no-priors* family is correct for claim–evidence factuality, where the source passage’s own truth is at stake, and the *with-priors* family is what coherence requires. The two symmetric couplings EQUIVALENCE and EXCLUSIVE coincide in both families, having no source-prior term to correct.

C.2 THE FOUR READOUTS

Definition 1 specifies a distribution; a coherence score is a functional of it, and several are defensible. We state five criteria, define four candidates with their exact constructions, and then decide between them by exact inference rather than by preference.

Definition 3 (Criteria). *A coherence readout should be: (R1) a genuine functional of the joint $\mathbb{P}(\mathbf{a})$, not of the relation list alone; (R2) valued in $[0, 1]$ and interpretable; (R3) monotone — resolving or removing a conflict must not decrease it; (R4) sensitive to conflict without hand-tuned constants; (R5) computable from standard inference queries.*

R1 deserves comment, since it is the criterion that does the most work. A readout that depends only on $\mathcal{R}$ — how many relations there are, what types, what weights — is not measuring the response; it is measuring its own annotation. Proposition 3 below shows this is not a hypothetical concern.

C.3 (A) MEAN POSTERIOR MARGINAL

$$\text{LCS}_{\text{mm}}(y) = \frac{1}{n} \sum_{i=1}^n \mathbb{P}(a_i = 1). \quad (4)$$

The marginals come directly from a single MAR query. The interpretation is the expected fraction of the response’s claims that the argument, taken as a whole, upholds. A conflict suppresses the marginals of its endpoints; a satisfied entailment chain propagates support along itself; a claim no relation touches stays at its prior and contributes neutrally.

Alongside the scalar we report three diagnostics that come free with the same query: the per-claim marginals themselves, which localize an incoherence to specific claims; the count of claims dragged *below their own prior*, which is the sharpest single indicator of an active conflict; and the mean normalized binary entropy $\mathcal{E} = \frac{1}{n} \sum_i H_2(q_i)$, lower being more decisive.

C.4 (B) CONSISTENCY: A TWO-TERM PROBABILITY

The obvious alternative to averaging marginals is to ask for the probability that the response is consistent. Making that precise turns out to require two terms:

$$\text{LCS}_{\text{cn}} = \frac{1}{2}(\mathcal{K} + \mathcal{S}), \quad \mathcal{K} = \mathbb{P}\left(\bigwedge_{r: \tau_r = \text{CONTRADICTION}} \neg(a_{s_r} \wedge a_{t_r})\right), \quad \mathcal{S} = \frac{\sum_{r \in \mathcal{R}_S} p_r U_r}{\sum_{r \in \mathcal{R}_S} p_r}, \quad (5)$$

where $\mathcal{R}_S$ ranges over $\{\text{ENTAILMENT}, \text{EQUIVALENCE}, \text{EXCLUSIVE}\}$ and U_r is the probability that relation r is *actively upheld*:

$$U_r = \mathbb{P}(a_{s_r}=1 \wedge a_{t_r}=1) \text{ for ENTAILMENT, EQUIVALENCE,} \quad U_r = \mathbb{P}(a_{s_r} \neq a_{t_r}) \text{ for EXCLUSIVE.}$$

The two coupling sets are deliberately different. The conflict event $\mathcal{K}$ keys on CONTRADICTION alone, even though $\mathcal{T}_{\text{conf}} = \{\text{CONTRADICTION}, \text{EXCLUSIVE}\}$. The reason is that an EXCLUSIVE spreads its incoherence over *both* same-value worlds, so reading only its both-true half would describe half the coupling and silently report the other half as satisfactory. EXCLUSIVE therefore earns credit in $\mathcal{S}$ instead, where its exactly-one world is simultaneously where the exclusion is honoured and where it is informative. CO_NECESSITY enters neither term: its defect is the both-false world, which (a)’s marginals already see directly.

Computing $\mathcal{K}$ exactly. $\mathcal{K}$ is an event probability over a conjunction of $|\{r : \tau_r = \text{CONTRADICTION}\}|$ pairwise events, which a naive encoding would represent as a factor of exponential arity. Instead we augment the network with deterministic auxiliary variables: one u_r per conflict edge with a ternary factor enforcing $u_r = a_{s_r} \wedge a_{t_r}$, then a chained running disjunction $U_k = U_{k-1} \vee u_k$ with ternary factors, and finally $\mathcal{K} = \mathbb{P}(U_{|\cdot|}=0)$. Every auxiliary factor is deterministic and at most ternary, so no $2^{|\mathcal{R}|}$ table is ever built and the augmented network is read with one MAR.

Computing $\mathcal{S}$ exactly. $\mathcal{S}$ is not an event probability but a weighted sum of *pairwise joints* — a linear functional of the joint — so it cannot be read off the same accumulator chain. It needs one deterministic indicator variable per supported edge and its own MAR. Because the indicators are mutually independent given the base network, they may be batched across several queries without approximation, which matters only for inference backends that enumerate.

Why “actively upheld” and not “satisfied”. The definition of U_r credits a relation only in the informative world, not in every world where it is technically satisfied. An entailment $a_s \Rightarrow a_t$ is satisfied vacuously whenever $a_s = 0$, and crediting that is fatal:

Proposition 3 (The vacuous-truth trap). *Let $\mathcal{K}' = \mathbb{P}(\text{every relation in } \mathcal{R} \text{ is satisfied})$, the apparently natural extension of $\mathcal{K}$ to all couplings. Then a response with no mined relations satisfies $\mathcal{K}' = 1$, the maximum attainable value.*

Proof. $\mathcal{K}'$ is the probability of an intersection of $|\mathcal{R}|$ events; for $\mathcal{R} = \emptyset$ the intersection is the whole space. $\square$

The proposition is immediate, but its consequence is not, and it is the reason LCS_{cn} has the form it does. Measured on the family of §F.2, a disconnected bag of claims scores $\mathcal{K}' = 1.000$ while a response whose causal spine is fully satisfied scores 0.221 — an inversion, not a tie. Nor is the defect repaired by count normalization: the geometric mean over relations gives 0.883 for the base response against 0.882 for the coherent one, non-monotone *and* still 1.000 on the bag; the expected fraction of satisfied relations gives 0.857 against 0.852, likewise inverted; and normalizing against a prior ceiling leaves $[0, 1]$ altogether (8.82 against 5.46) while still running backwards. The common cause is that “are the asserted relations honoured?” is a property of $\mathcal{R}$ rather than of y — a violation of R1 — and it is maximized by asserting nothing. Crediting only the informative world removes the free lunch: a bag of claims upholds nothing and scores $\mathcal{S} = 0$.

The two terms are averaged rather than multiplied so that neither can annihilate the other. A live contradiction should not erase a well-supported spine, and a relation-free response should not inherit a perfect score from an empty conflict set.

C.5 (C) A REIFIED COHERENCE NODE

The most direct reading of “is this response coherent?” as a single probability is to introduce a variable for it. Add a binary R with Bernoulli prior ρ and, for each relation, a ternary *vote* factor

$$h_r(R, a_s, a_t) = \begin{cases} 1 & R = 0 \quad (\text{the branch is flat}) \\ 1 & R = 1, (a_s, a_t) \notin V(\tau_r) \\ 1 - p_r & R = 1, (a_s, a_t) \in V(\tau_r), \end{cases} \quad \text{LCS}_{\text{re}} = \mathbb{P}(R = 1). \quad (6)$$

In the $R = 1$ branch each violated relation charges $1 - p_r$, so R behaves as a noisy-AND over relation satisfaction; in the $R = 0$ branch nothing is charged, so $R = 0$ carries no commitment. With no relations R is decoupled and $\text{LCS}_{\text{re}} = \rho$, correctly signalling that nothing was measured. Note that the violated set is $V(\tau)$ from Def. 2, so ENTAILMENT is treated as satisfied at $a_s = 0$ — vacuously — which is where (c) inherits a weaker form of Prop. 3’s problem. Appendix I works a three-claim subnet through by hand, giving $\mathbb{P}(R=1) = 0.453$.

C.6 (D) NORMALIZED LOG-PARTITION

The partition function measures how much probability mass the constraints leave standing, which is a global coherence signal requiring no marginals at all. It is unnormalized, so we grade it between two references built from the same edge skeleton:

$$\text{LCS}_{\text{lp}} = \frac{\log Z - \log Z_{\min}}{\log Z_{\max} - \log Z_{\min}}. \quad (7)$$

For the ceiling, $Z_{\max}$ is the partition function of the same network with every conflict edge removed: deleting constraint factors only adds mass, so $\log Z \leq \log Z_{\max}$ always. For the floor we take the MAP world mass of the base network, $Z_{\min} = \max_{\mathbf{a}} \prod_i \phi_i \prod_r \psi_r$.

Lemma 1 (The MAP world mass is a valid, coherence-graded floor). $\log Z_{\min} \leq \log Z$ for every network, and $Z_{\min}$ is non-increasing in conflict strength.

Proof. $Z = \sum_{\mathbf{a}} \text{mass}(\mathbf{a})$ is a sum of non-negative terms, so its single largest term is a lower bound. For the second claim, increasing p_r on a conflict edge multiplies the mass of every world in $V(\tau_r)$ by a factor below one and leaves other worlds unchanged, so the maximum over worlds cannot increase. $\square$

Lemma 1 is not a technicality; it replaces two floors that look more natural and are wrong. Retyping every edge to CONTRADICTION, or saturating conflicts to $p = 1$, both fail to bound $\log Z$ below for the row-stochastic tables of Prop. 1: an isolated conflict removes no mass at all under those tables, whereas a mass-concentrating entailment backbone concentrates a great deal, so a network combining a strong backbone with several soft conflicts can have a base $\log Z$ below either candidate floor. We observed exactly this on real graphs, which put Eq. 7 outside $[0, 1]$ until the MAP floor replaced them. In practice we clamp the ratio to $[0, 1]$ regardless, since approximate inference computes $\log Z$ and the MAP bound with finite i-bound and small tolerance violations are possible.

C.7 DECIDING BETWEEN THE FOUR

Table 8 settles the question empirically and Table 9 summarizes. Readouts (b) and (c) both fail R3, and they fail it at the *same* variant — the one where a conceded tension is resolved — for a shared reason that is worth stating precisely, because it looks like an implementation bug and is not.

Proposition 4 (Belief versus event activity). *Softening a resolved conflict via Eq. 2 increases the probability of its both-true world. For a single conflict edge with uniform priors, discounting p from 0.80 to 0.55 moves $\mathbb{P}(a_s=1, a_t=1)$ from 0.10 to 0.225.*

Proof. Under both conflict couplings the $(1, 1)$ cell of Table 7 is $1 - p$, which increases as p falls. Take a lone CONTRADICTION edge with $\pi_s = \pi_t = \frac{1}{2}$, so the with-priors table is $[\frac{1}{2}, \frac{1}{2}, p, 1 - p]$ and each world carries an equal unary factor of $\frac{1}{4}$. The unnormalized masses are then proportional to the table itself, giving $\mathbb{P}(1, 1) = (1 - p) / (\frac{1}{2} + \frac{1}{2} + p + (1 - p)) = (1 - p) / 2$, which is monotone decreasing in p . Evaluating at $p = 0.80$ and $p = 0.55$ gives 0.100 and 0.225. $\square$

Table 8: All four readouts across a coherence-ordered family of five variants of one response, by exact inference over 2^{16} worlds. The variants run from least to most coherent left to right in the intended ordering. Bold marks a value that moves the *wrong* way relative to its predecessor. Only (a) and (d) are monotone throughout.

Variant	(a) LCS_{mm}	(b) LCS_{cn}	(c) LCS_{re}	$\log Z$	(d) LCS_{lp}
<i>worse</i>	0.586	0.778	0.116	-10.54	0.68
<i>base</i>	0.607	0.781	0.148	-9.64	0.78
<i>concession</i>	0.612	0.772	0.134	-9.64	0.80
<i>fixcasualty</i>	0.613	0.764	0.156	-8.95	0.89
<i>coherent</i>	0.620	0.752	0.181	-8.25	1.00
monotone?	yes	no	no	yes	yes

Table 9: The four readouts against the criteria of Def. 3, with the inference queries each requires.

Readout	Queries	R1	R2	R3	R4	R5	Note
(a) mean marginal	MAR	✓	✓	✓	✓	✓	selected
(b) consistency	$MAR \times 2$	✓	✓	×	✓	✓	concession dip
(c) reified	MAR	✓	✓	×	×	✓	free p ; dip
(d) norm. $\log Z$	PR, MAP	✓	✓	✓	✓	✓	needs 2 references

The consequence sorts the four readouts into two kinds. A readout that asks *is the conflict active* — (b) counts exactly that event, (c) charges $1 - p_r$ for it — must register a softened conflict as *more* often active, and therefore dips. A readout that asks *how strongly should each claim be believed* — (a) through marginals, (d) through mass — sees less suppression and rises. Since resolving a concession is by construction an increase in coherence, a coherence score must track belief rather than event activity. That is the substantive finding of this section, and it is not specific to our parameterization: any readout defined as the probability of a conflict-free configuration will invert under any discount-based treatment of resolved concessions.

A second failure mode of (b). Table 8 shows (b) failing worse than a single dip, and the reason is instructive about $\mathcal{K}$ and $\mathcal{S}$ jointly. In this family both conflicts are EXCLUSIVE rather than CONTRADICTION, so $\mathcal{K} \equiv 1.000$ throughout by the coupling-set decision above, and the score is carried entirely by $\mathcal{S}$. But removing an exclusion removes *support* credit as well as conflict, so the score runs backwards from 0.778 to 0.752 across the whole ladder. On three-coupling variants of the same response, where the conflicts are CONTRADICTION and $\mathcal{K}$ is live, (b) instead runs $0.576 \rightarrow 0.654 \rightarrow \mathbf{0.582} \rightarrow 0.672 \rightarrow 0.752$: correctly ordered apart from the concession dip. Either way it is non-monotone, and the lesson is that $\mathcal{K}$ and $\mathcal{S}$ should be read separately as diagnostics rather than averaged into a headline.

Selection. We report (a) as the headline. It satisfies all five criteria, needs a single MAR query, and is the only one of the four whose value decomposes into per-claim contributions — which is what makes an incoherence *locatable* rather than merely detectable. We compute and report all four, since (d) is a genuine alternative satisfying the same criteria through a different route (mass rather than marginals), and $\mathcal{K}$, $\mathcal{S}$ carry diagnostic information the mean marginal does not expose.

C.8 ALGORITHM AND COST

Algorithm 1 is dominated by steps 4–9: $|P|$ Prompt-A calls plus one Prompt-B call per surviving edge, which the candidate policy keeps near-linear in n rather than quadratic. Inference is negligible by comparison.

Answering all four readouts requires seven inference calls: the base MAR and PR, the $\mathcal{K}$ accumulator MAR, the $\mathcal{S}$ indicator MAR, the reified MAR, the conflict-free ceiling PR, and the base MAP. Computed independently they would cost fourteen, since each readout re-runs the shared base pair. Production inference is weighted mini-bucket (Liu & Ihler, 2011) with i-bound 6; every *exact* value reported

Algorithm 1 COMPUTELCS(y) — the end-to-end coherence score.

Require: response y ; relation estimator $\mathcal{M}$; discount λ ; priors π
Ensure: LCS $\in [0, 1]$, marginals $\{q_i\}$, diagnostics

- 1: $\mathcal{A} \leftarrow \text{ATOMIZEANDDECONTEXTUALIZE}(y)$
- 2: $P \leftarrow \text{SELECTCANDIDATEPAIRS}(\mathcal{A}, y, \dots)$ ▷ Algorithm 2
- 3: $\mathcal{R} \leftarrow \emptyset$
- 4: **for all** $(a_i, a_j) \in P$ **do** ▷ concurrent, rate-limited
- 5: $(\text{sense}, \tau, c) \leftarrow \text{PROMPTA}(a_i, a_j, y)$ ▷ $c = \mathbb{P}(\tau \mid \cdot)$ from Eq. 3
- 6: **if** $\tau = \text{NONE}$ **then continue**
- 7: $\sigma \leftarrow \text{PROMPTB}(a_i, a_j, \tau, y)$ ▷ surrogate-token strength
- 8: $\mathcal{R} \leftarrow \mathcal{R} \cup \{(i, j, \tau, \text{clip}_{[0,1]}(c \cdot \sigma))\}$
- 9: **for all** $r \in \mathcal{R}$ with $\tau_r \in \mathcal{T}_{\text{conf}}$ sensed as **Concession** **do**
- 10: **if** a resolving holding h exists **then** $p_r \leftarrow p_r(1 - \lambda\pi_h)$
- 11: **else** mark r unresolved
- 12: $\mathcal{N} \leftarrow$ network of Eq. 1 over $(\mathcal{A}, \mathcal{R}, \pi)$ ▷ with-priors tables
- 13: $\{q_i\} \leftarrow \text{INFER}(\mathcal{N}, \text{MAR})$
- 14: **return** $\frac{1}{n} \sum_i q_i, \{q_i\}, (\#\{q_i < \pi_i\}, \mathcal{E}, \log Z)$

Algorithm 2 SELECTCANDIDATEPAIRS: response-anchored candidate selection.

Require: claims $\mathcal{A} = \langle a_1, \dots, a_n \rangle$ in assertion order; response y ; policy; window w ; gate threshold θ_g ; discourse thresholds θ_d, δ
Ensure: ordered candidate pairs P , plus a coverage record

- 1: **if** policy = ALL_PAIRS **then return** every ordered pair $(a_i, a_j), i \neq j$
- 2: $W \leftarrow \{(a_i, a_j) : 0 < j - i \leq w\}$ ▷ forward order window
- 3: **if** policy = BIDIRECTIONAL **then** $W \leftarrow W \cup \{(a_j, a_i) : 0 < j - i \leq w\}$
- 4: $G \leftarrow \emptyset$
- 5: **if** policy = GATED **then**
- 6: $G \leftarrow \{(a_i, a_j) : j - i > w, \text{sim}(a_i, a_j) \geq \theta_g\}$
- 7: align each claim to its best-matching sentence of y
- 8: $\text{Prom} \leftarrow \{(a_i, a_j) \notin W \cup G : \text{DISCOURSEADJACENT}(i, j)\}$
- 9: $\text{Dem} \leftarrow \{(a_i, a_j) \in W : \neg \text{DISCOURSEADJACENT}(i, j)\}$
- 10: **return** $(\text{DEDUP}(W \cup G \cup \text{Prom}) \setminus \text{Dem}), (|W|, |G|, |\text{Prom}|, |\text{Dem}|)$

in this paper instead comes from explicit 2^n enumeration, which we use as a regression oracle for $n \leq 20$ and which is the source of Tables 8 and the values in §F.

D RELATION MINING IN DETAIL

D.1 CANDIDATE-PAIR SELECTION

Each claim is aligned to its originating sentence; a pair is *discourse-adjacent* if the two claims share content tokens above a Jaccard threshold, if their sentences are within a short span, or if the target’s sentence opens with a discourse connective. Out-of-window pairs that are discourse-adjacent are *promoted* into the candidate set; in-window pairs that are not are *demoted* out of it. The refinement is a correlated filter — it selects for the same surface adjacency that the sense prompt reads as evidence of relatedness — which is why we report it as a policy component and measure its effect rather than treating it as neutral preprocessing.

D.2 WHY THE EVIDENCE SPAN MUST QUOTE THE RESPONSE

The evidence-span requirement of §3.5 obliges the model to quote the words that draw the link from the *response*, not from either claim, and the parser then verifies the quotation is genuinely a substring of the response. Quoting the claims would not work: claims are decontextualized rewrites, and only

about 27% of them appear in the response verbatim even after normalization, so a claim-quoting rule would reject most genuine relations.

D.3 THE TWO PROMPT GENERATIONS

Two prompt generations. The first-generation Prompt A over-asserts badly: against a corpus in which 7.2% of candidate pairs are genuinely related, it answers “related” on 72% of the pairs a windowed policy hands it. Five properties push it that way, and the revision addresses each. (i) Its few-shot balance was three related to one unrelated; class balance in the exemplars is a strong prior on the answer distribution, and this one pointed the wrong way by an order of magnitude. The revision is two related to three unrelated. (ii) Its single negative exemplar had the response *explicitly disavow* a link (“the two processes were not related this year”), which teaches that NONE requires denial, whereas real prose simply stays silent; the revision’s negatives are all silence. (iii) It asked for grounding but gave the model nowhere to show it and nothing to check it — hence the evidence span. (iv) Its first reasoning step was a leading question listing eight affirmative relation types before mentioning NONE; the revision asks the discriminating question first, whether the text draws the link or the claims are merely near each other. (v) It gave NONE one line against nine for relation-bearing senses; the revision gives it first position and equal prose. The revision also names *transitivity* explicitly as a non-relation, since roughly 85% of the first generation’s false positives are the model closing a chain the text laid out pairwise, and 96.6% of them touch an endpoint of a real relation. Both generations are retained permanently so that any claim about the prompt can be measured rather than asserted; §4 reports both.

D.4 COUNTING FAILURES

Failures must be counted, not absorbed. Mining issues one LLM call per candidate pair for Prompt A and one more for each pair that yields an edge, run concurrently under a rate limit. A throttled or failed call returns no parse, and a parser that reads “no parse” as “no relation” converts an infrastructure failure into a sparse graph — which then scores as a *more* coherent response, since fewer conflicts are asserted. We therefore tally call errors separately from genuine NONE answers and report them, and distinguish an explicit NONE from an unparseable coupling for the same reason.

D.5 ALIGNING CLAIMS ACROSS THE TWO STAGES

Aligning claims across stages. The two stages must agree on which claim is which, and the obvious key is the wrong one. Matching proceeds by object identity first (the shared-atomization path, where the response is decomposed once and both stages see the same objects), then by normalized text (casefolded, whitespace-collapsed, terminal punctuation stripped), and only then by identifier. Text precedes identifier deliberately: near-duplicate removal keeps the first-seen claim and leaves the surviving identifier set *sparse* — $a_0, a_1, a_3, \dots$ — so two independent decompositions of the same response can disagree about which claim occupies which index. Matching on identifiers alone would then attach a prior to the *wrong* claim, which is strictly worse than attaching none, since it silently corrupts the score rather than degrading it. When coverage falls below one half the run is flagged degraded, and the configured policy is to warn, to raise, or to discard every prior and fall back to a clean coherence-only score rather than report a half-primed mixture.

E EVALUATION PROTOCOL AND BASELINES IN FULL

E.1 THE FIVE BASELINE FAMILIES

A coherence measure should be compared against controls chosen to fail in informative ways. We describe five families and state what each is diagnostic of. Four are implemented and share the LCS’s own decomposition — each baseline is handed the same atoms the LCS was scored on and, like the coherence MRF, consults no retrieved evidence — so the comparison is an ablation rather than a separate experiment. Metrics requiring a source document or a reference are therefore excluded by construction, and we say so where the exclusion is not obvious.

1. *Ladder-blind controls*: claim count, relation count, response length, and number of perturbation operators applied. These cost nothing and they catch the most likely way a coherence measure is secretly wrong — by tracking edit count or graph density rather than coherence. A measure that cannot beat them on C3, or that fails to stay flat where they do, has not earned its complexity. Steen & Markert (2022) make the same argument for coherence measures generally, and introduce intra-system correlation and bias matrices to identify measures whose apparent performance comes from system-level confounders; we adopt the controls in that spirit. A control that *succeeds* is bad news for the item, not good news for the control, and §E reports one such case.
2. *Local contradiction detection*: pairwise NLI over claim pairs with no global model, scored as one minus the fraction of pairs labelled contradictory, in the spirit of SelfCheckGPT’s NLI variant (Manakul et al., 2023). This isolates precisely what joint inference adds over local detection. The prediction is that it does creditably where the defect is a *present* contradiction and poorly where the defect is a *missing* entailment, since a broken chain has no contradictory pair to find. We run it with the same NLI extractor as §A, over the same atoms, so the only differences from the LCS are relation *typing* and *propagation*.
3. *Structured reasoning consistency*: ROSCOE’s Self-Consistency (Golovneva et al., 2023), $1 - \max_i \max_{j < i} p_{\text{contr}}(h_i, h_j)$, which is evidence-free like ours but differs in three separable ways — it is a *maximum*, so a second conflict changes nothing; it is *forward-only*, so a relation running backward in claim order is unreachable (§3.5 measures 105 of 603 gold relations as both directed and backward); and it is *untyped*. We therefore also report a mean-aggregated and a symmetric variant, which turn one baseline into an ablation identifying *which* of the three differences accounts for any gap. ROSCOE targets step-by-step reasoning chains, and a long-form response is not a chain, so we label it adapted.
4. *LLM judges*: G-Eval’s coherence dimension (Liu et al., 2023), with the SummEval coherence rubric (Fabbri et al., 2021), and a direct “rate the logical coherence from 1 to 5” prompt as the no-rubric control. These are the practical incumbent and the comparison reviewers will want. The ladder is a demanding test for them, since consecutive rungs differ by a few words, so a judge must be sensitive at very small edit distances while remaining flat where meaning is preserved. We report each judge as a mean over five runs with its standard deviation, because the determinism of Remark 2 is only an advantage next to a measurement of the judge’s instability, and because a spread wider than the gap being discussed bounds what the comparison can conclude.
5. *Discourse representations*: DiscoScore (Zhao et al., 2023), with an entity-grid model (Barzilay & Lapata, 2008) as the classical reference. DiscoScore is presented as reference-based and most of its variants consume a reference, which a standalone response does not have; three of them (RC, LC and ENTITYGRAPH) ignore the reference argument entirely and score a single document, and those are the three we use — the last of which *is* an entity grid, so the classical reference comes from the same implementation. All three reduce to noun-lemma repetition across sentences, with no representation of entailment or contradiction, which makes them a sharp test of whether coherence is separable from *cohesion* rather than a weak competitor: §E reports a pair of responses with identical nouns and identical repetition structure, one self-consistent and one self-contradictory, on which all three are exactly flat.
6. *Internal ablations*: uniform versus factuality priors (§3.6); the four readouts of §C; the four candidate-pair policies; the two Prompt-A generations; and the gold-relation ceiling. These separate the model’s design choices from the estimator’s accuracy.

Beyond ladder ordering we recommend Spearman correlation against human coherence ratings on the same items, reported alongside per-family ordering agreement. Ordering agreement is the sharper instrument, being a controlled contrast with a known answer; human correlation is what establishes that the construct being ordered is the one readers care about. Neither substitutes for the other.

A baseline we cite but do not run. The closest existing work to our relation-mining stage is Liu & Strube (2025), who also build a graph over sentence pairs labelled with PDTB 3.0 senses — via a discourse parser rather than an LLM — and add entity links. We do not report a head-to-head number, and the reason is a difference in output type rather than a convenience: their model emits a *three-way* label (low, medium, high), and a three-way label cannot be checked against the ordering constraints of §4.1, which ask whether consecutive rungs move in a declared direction. The comparison would also require their trained classifier and a PDTB parser, neither of which is a pretrained scorer one

can apply to a new response. Three differences are worth recording regardless, because they are the axes on which the two approaches diverge: their model performs no probabilistic inference, so a local conflict cannot propagate; it retains only forward-order pairs ($i < j$), so a backward relation is unreachable, which is the same structural cap we measure in §3.5; and it produces no per-claim attribution, so an incoherence is detected but not located.

E.2 WHAT THE BASELINES SHOW BEFORE ANY LADDER SWEEP

Two properties of these baselines are worth establishing before any ladder sweep, because both bear on what the comparison in §4.4 can support.

First, *cohesion is not coherence, and the discourse floor cannot tell the difference*. On a pair of two-sentence responses with identical nouns and identical repetition structure — “The weld passed the tensile test. The weld held under load.” against “... The weld *failed* under load.” — all three single-document DiscoScore metrics return *exactly* equal values, while the pairwise-NLI and ROSCOE baselines separate the pair. The metrics are defined over noun repetition, so this is a property of their definitions rather than an accuracy failure; it is also the cleanest available demonstration that the axis we measure is not the one entity-based coherence models measure.

Second, *a controlled contrast has to be checked, not assumed*. A measure can reproduce a declared ordering for reasons that have nothing to do with coherence, so a contrast is only informative once the model-free proxies have been shown *not* to reproduce it. The claim-identical ordering pair of §F.3 is the clean case: claim count is exactly flat there by construction, response length differs by four tokens (265 against 269) in the direction *opposite* to the declared ordering, and all three discourse metrics rank the adversarial summary *above* the faithful one, the self-serving-first ordering carrying more noun repetition. A cohesion metric prefers the less coherent response of the two. The ladder corpus of §4 generalizes exactly this property: every rung of a family shares one claim set, so a claim-count proxy is flat by construction across all seventy items.

E.3 MEAN READOUT VALUES PER ARM

Table 2 of §4 reports only the constraint counts, since averaging a readout across the rungs of a ladder collapses the ordering the ladder exists to test. The mean values are recorded here as a level comparison. The BIDIRECTIONAL density cost is visible quantitatively: LCS_{lp} collapses to 0.037 and 0.033 against gold’s 0.410 on the same seventy responses, from adding roughly two thousand spurious factors — the log-partition readout registers added factor mass most directly.

Arm	Policy	LCS_{mm}	LCS_{cn}	LCS_{re}	LCS_{lp}
GOLD	—	0.795	0.515	0.238	0.410
GOLD_VALID	—	0.794	0.576	0.316	0.485
<i>mined by llama-3.3-70b</i>					
	WINDOWED	0.796	0.676	0.384	0.224
	BIDIRECTIONAL	0.723	0.542	0.205	0.037
<i>mined by gpt-oss-120b</i>					
	WINDOWED	0.825	0.500	0.425	0.129
	BIDIRECTIONAL	0.808	0.462	0.251	0.033

Two further measurements belong with these. The cohesion baselines drift by at most 0.0096 across a permuted family, which is why they score 16/20 rather than 20/20 on invariance. And the 0.014 run-to-run F1 movement of §4.1 supersedes the ± 0.06 we previously estimated on a ten-item corpus: the larger corpus tightens reproducibility rather than merely enlarging the denominator.

E.4 JUDGE SATURATION, MEASURED

The two judges of §4.4 fail in opposite ways, and the mechanism is saturation rather than noise. llama-3.3-70b’s G-Eval is astonishingly stable and useless: mean per-item standard deviation 0.0002, only three distinct values across the seventy items, and a mean score of 0.996 — it rates almost everything at ceiling, so it cannot order anything. gpt-oss-120b’s G-Eval uses its range (fourteen distinct values, mean 0.749) and pays in stability (mean sd 0.0800, max 0.187). Reporting

Table 10: Declared ordering assertions satisfied on the ladder corpus, under two different judge models. Only the readout-independent content of a constraint is scorable by a single-score baseline: the four C1 consecutive-increase pairs and the C3 endpoint separation, giving five per family. C2 is excluded because it predicts the internal behaviour of particular readouts (§C.7), which a baseline makes no claim about. The ORDER and CONTROL families assert invariance rather than increase and so contribute *no* scorable assertion, leaving ten families and 50 assertions; the LCS rows are scored on *exactly* those 50, so they are comparable to the rows below them, while the 82.7% of Table 2 is computed over the fuller 202-assertion set including invariance and is not. Judges are the mean of five runs. Model-free baselines are identical under both judges, as they must be; the two judge rows are the entire difference.

Measure	Assertions satisfied				Uses LLM
	llama-3.3-70b		gpt-oss-120b		
LCS _{mm} (gold relations)	37/50	74.0			no
LCS _{lp} (gold relations)	36/50	72.0			no
LCS _{cn} (gold relations)	34/50	68.0			no
RC (cohesion)	30/50	60.0	30/50	60.0	no
LC (cohesion)	28/50	56.0	28/50	56.0	no
LENGTH (control)	20/50	40.0	20/50	40.0	no
ROSCOE-SC (mean)	16/50	32.0	17/50	34.0	yes
CONTRADICTION RATE	12/50	24.0	12/50	24.0	yes
JUDGE, direct rating	4/50	8.0	18/50	36.0	yes
JUDGE, G-Eval	0/50	0.0	21/50	42.0	yes
ROSCOE-SC (max)	0/50	0.0	0/50	0.0	yes
ENTITY GRAPH	0/50	0.0	0/50	0.0	no
CLAIM COUNT (control)	0/50	0.0	0/50	0.0	no

judge variance alone would have called the first judge the reliable one, which is exactly backwards for the task. Both judges also violate invariance freely: gpt-oss’s G-Eval moves by up to 0.55 across rungs that differ only in sentence order.

E.5 PER-MEASURE LADDER AGREEMENT

F WORKED EXAMPLES IN FULL

We work four examples. The first is the smallest complete instance of the two-stage model: one claim set, two responses, exact inference. The second is a richer sixteen-claim graph that exercises the conflict taxonomy. The third is a claim-identical pair that isolates ordering. The first three are hand-authored, which bounds what they can show: they demonstrate that the model reads a graph correctly, not that the estimator finds such graphs in text it has never seen. The fourth is therefore a corpus rather than an example — one hundred machine-generated responses whose incoherence is planted by construction, scored end to end with no hand-drawn relations anywhere.

F.1 A FIVE-CLAIM PAIR, END TO END

The point of this example is that factuality and coherence are separable, so we hold the first fixed and vary only the second. Five claims about the Voyager 1 mission are fixed once and for all, three of them true of the world and two false; stage one is taken as given, contributing the priors $\pi = 0.9$ for each true claim and $\pi = 0.1$ for each false one through the composition of §3.6:

	π_i	Claim
a_1	true 0.9	Voyager 1 crossed the heliopause in August 2012.
a_2	true 0.9	After the crossing, Voyager 1’s instruments measured a sharp rise in plasma density.
a_3	true 0.9	Voyager 1 is therefore in interstellar space.
a_4	false 0.1	Voyager 1’s plasma wave instrument failed permanently before the crossing.
a_5	false 0.1	Voyager 1 has left the Solar System entirely.

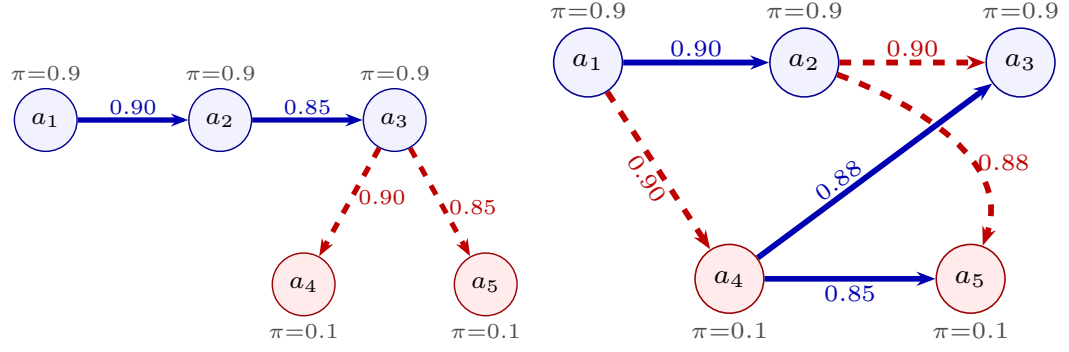


Figure 3: Relation graphs for the five-claim worked pair (§A). Blue nodes are claims true of the world ($\pi_i = 0.9$), red nodes false ($\pi_i = 0.1$); the *same* five claims appear in both panels. Solid blue: ENTAILMENT; dashed red: CONTRADICTION. Line width grows with the relation weight p . The panels differ only in *how the response arranges* the claims, which is the whole of what coherence measures — no per-claim factuality check can separate them.

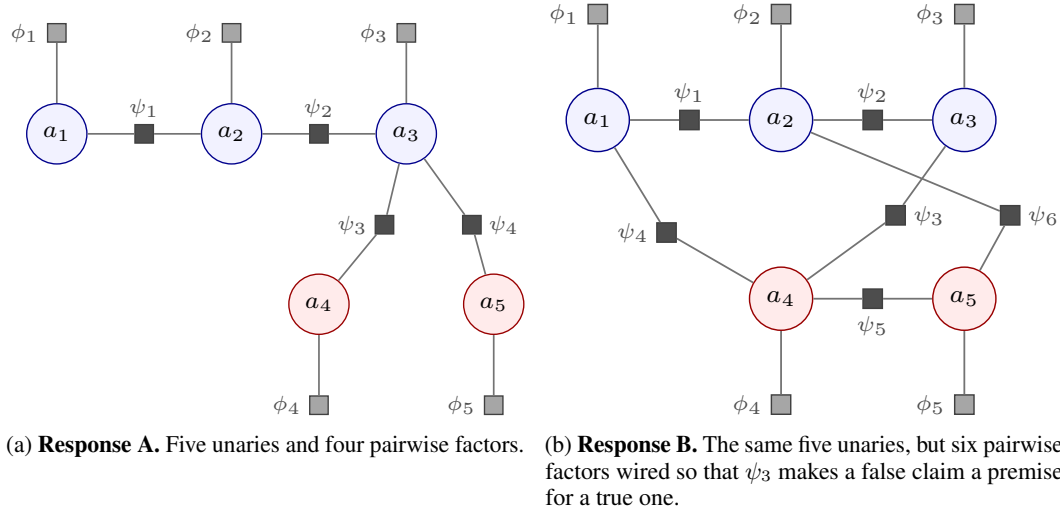


Figure 4: The coherence MRF of Eq. 1 for the two responses, as an explicit factor graph. Circles are the binary claim variables; grey squares the unary factors $\phi_i = [1 - \pi_i, \pi_i]$ that carry the priors; black squares the pairwise factors ψ_{τ_r, p_r} of Table 7, one per mined relation. **The priors enter only through the ϕ_i :** the claim-level potentials are identical across the two panels, and every difference in score comes from the ψ_r — their number, their endpoints, and their couplings.

Two responses then assert *these same five claims* and differ only in how they arrange them. Because the claim set is identical, every per-claim factuality check returns the same verdict on both; the relation graphs (Figure 3) and the MRFs they induce (Figure 4) are what differ. Table 11 gives both responses in full, with the span asserting each claim marked $[a_i]$ at its first mention.

The italicized spans are the ones the estimator reads as relations, and they are the whole difference between the two. Table 12 lays the two relation sets side by side, one row per claim pair, with each relation’s discourse sense and the coupling it compiles to. In A the false claims are named, attributed, and then refused on a stated ground, which compiles to a CONTRADICTION edge from a_3 .

Response A (coherent) — $\text{LCS}_{\text{mm}} = 0.590$, $\text{LCS}_{\text{cn}} = 0.963$, $\log Z = -0.95$

Voyager 1 crossed the heliopause in August 2012. *[a₁]* After the crossing, its instruments measured a sharp rise in plasma density, *[a₂]* and that measurement is what established the boundary had been passed; Voyager 1 is therefore in interstellar space. *[a₃]* Two claims that circulate about the mission do not survive scrutiny. It is sometimes said that the plasma wave instrument had failed permanently before the crossing, *[a₄]* *but it cannot have failed and also have returned the density measurement that settled the question.* It is also said that the probe has left the Solar System entirely; *[a₅]* *being in interstellar space is not the same thing, since the Oort cloud remains gravitationally bound and lies far beyond Voyager 1.*

Response B (incoherent) — $\text{LCS}_{\text{mm}} = 0.494$, $\text{LCS}_{\text{cn}} = 0.414$, $\log Z = -3.78$

Voyager 1 crossed the heliopause in August 2012, *[a₁]* and after the crossing its instruments measured a sharp rise in plasma density. *[a₂]* *That said, the probe cannot really be counted as being in interstellar space.* *[a₃]* The plasma wave instrument had in fact failed permanently before the crossing, *[a₄]* *and it is precisely that failure which tells us the probe reached interstellar space.* The instrument failure also means Voyager 1 has left the Solar System entirely. *[a₅]* *Of course, the August 2012 crossing rules out any such permanent failure, and the density measurement is flatly inconsistent with the probe having left the Solar System.*

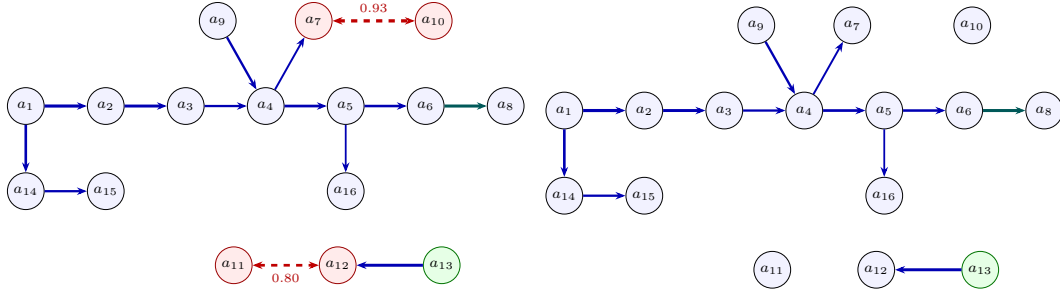
Table 11: The two responses in full. They assert the *same* five claims of §F.1; each claim’s first mention is marked *[a_i]*. Italicized spans are the ones the estimator reads as relations. In A they discharge the two false claims; in B they reason *from* them.

Table 12: The mined relations of both responses, one row per claim pair. *s* is the Level-2 discourse sense, τ the Level-1 coupling it compiles to (Table 5), p the weight. E abbreviates ENTAILMENT, C CONTRADICTION. These are exactly the edges of Figure 3 and the factors ψ_r of Figure 4; a dash means the response draws no relation over that pair.

Pair	Response A (coherent)				Response B (incoherent)				What differs
	Edge	s	τ	p	Edge	s	τ	p	
$\{a_1, a_2\}$	$a_1 \rightarrow a_2$	Cau-Eff	E	.90	$a_1 \rightarrow a_2$	Cau-Eff	E	.90	The one relation both agree on.
$\{a_2, a_3\}$	$a_2 \rightarrow a_3$	Evidence	E	.85	$a_2 \rightarrow a_3$	Contrast	C	.90	A supports its conclusion; B denies it. Support becomes conflict.
$\{a_3, a_4\}$	$a_3 \rightarrow a_4$	Contrast	C	.90	$a_4 \rightarrow a_3$	Evidence	E	.88	The arrow reverses. A rejects a_4 ; B makes it a premise.
$\{a_3, a_5\}$	$a_3 \rightarrow a_5$	Contrast	C	.85	—	—	—	—	A refuses a_5 ; B never draws the distinction.
$\{a_4, a_5\}$	—	—	—	—	$a_4 \rightarrow a_5$	Cau-Eff	E	.85	B alone chains one falsehood into the other.
$\{a_1, a_4\}$	—	—	—	—	$a_1 \rightarrow a_4$	Contrast	C	.90	B denies a_4 , which it had just used as a premise — this strands a_3 .
$\{a_2, a_5\}$	—	—	—	—	$a_2 \rightarrow a_5$	Contrast	C	.88	Likewise for a_5 : asserted, then denied.
Totals		2 E, 2 C				3 E, 3 C			A’s conflicts point <i>outward</i> , at the false claims; half of B’s point <i>inward</i> .

In B the same falsehoods are named and then *used* — a_4 is offered as the reason a_3 holds — while the response also denies a_3 outright and, in its last sentence, denies a_4 and a_5 too. Mentioning a falsehood is not what costs B its score; asserting it and reasoning from it is. All values below are exact, by $2^5 = 32$ -world enumeration.

Example 2 (Coherent). *Response A’s four relations (Table 12, column A) are a two-edge support chain over the true claims and one rejection of each false claim. The model returns $\text{LCS}_{\text{mm}} = 0.590$ and $\text{LCS}_{\text{cn}} = 0.963$ with $\log Z = -0.95$. Every claim finishes on the correct side of its own prior: the true claims sit at 0.939, 0.989 and 0.992 — each above its 0.9 prior; because the chain supplies mutual support — and the false ones at 0.013 and 0.020, pushed below 0.1 by the rejections. Nothing in the response fights anything else, so no claim is penalized.*



(a) **Incoherent** (base). Two EXCLUSIVE conflicts (dashed): $a_7 \leftrightarrow a_{10}$ is *unresolved*; $a_{11} \leftrightarrow a_{12}$ is a concession the holding a_{13} (green) resolves. $\text{mm} = 0.607$, $\log Z = -9.64$. (b) **Coherent**. The identical support structure with both conflict edges removed; a_{10} and a_7 are no longer suppressed. $\text{mm} = 0.620$, $\log Z = -8.25$, $\text{lp} = 1.00$.

Figure 5: Relation graphs for the AeroParts recall report (§A). Solid blue: ENTAILMENT; teal: EQUIVALENCE; dashed red, double-headed: EXCLUSIVE. Line width grows with the relation weight p . The two panels differ only in the two conflict edges, so the $0.607 \rightarrow 0.620$ lift and the 1.39 nats of recovered mass are attributable to conflict structure alone.

Example 3 (Incoherent). *Response B keeps only $a_1 \rightarrow a_2$ and replaces the rest with six relations that pull against each other (Table 12, column B): it denies its own conclusion, promotes the false a_4 to a premise for that same conclusion, chains a_4 into a_5 , and then denies both falsehoods in its closing sentence. Three of its six relations are conflicts, and two of those point at claims the response itself asserts. The score falls to $\text{LCS}_{\text{mm}} = 0.494$ and $\text{LCS}_{\text{cn}} = 0.414$, with $\log Z = -3.78$. The diagnostic that localizes the damage is *per-claim*: the true a_3 is dragged to 0.513, far below its own 0.9 prior, because its only support arrives from a_4 , which the response itself denies — and a premise the text refuses cannot transmit support. Three claims finish below their own priors, against two in Response A, and a_3 is the one that matters: a claim true of the world, demoted to a coin flip by the argument built around it.*

Two things are worth reading off this pair. First, the priors enter *only* through the unary factors ϕ_i of Figure 4, which are identical in both panels; the entire $0.590 \rightarrow 0.494$ drop is produced by the ψ_r , so it is attributable to arrangement alone. Second, a claim being true of the world does not protect it: coherence is a property of the joint distribution, and a_3 is suppressed by *how* it is argued for, not by what it says. This is the mechanism Remark 2 anticipates and the ordering pair of §F.3 exhibits at scale.

A caution on readouts. LCS_{lp} is 0.194 for Response A and 0.267 for Response B, which inverts the order — correctly, and for a reason Eq. 7 makes plain: its references $\log Z_{\text{max}}$ and $\log Z_{\text{min}}$ are recomputed *per response* (-0.49 and -1.06 for A, -1.65 and -4.55 for B), so it reports how close a response comes to the ceiling of *its own* edge skeleton, not how the two responses compare. It is a within-response diagnostic. Raw $\log Z$ is comparable across responses and moves the right way, by 2.83 nats.

F.2 A SIXTEEN-CLAIM RECALL REPORT

The second example is a sixteen-claim aviation recall report, which exercises the conflict taxonomy more fully than the five-claim pair: it carries both an unresolved conflict and a conceded one. Its relation graph is twelve support edges — an entailment spine $a_1 \rightarrow a_2 \rightarrow \dots \rightarrow a_6$ (shipment certified to spec, installation, entry into service, in-flight failures, regulatory probe, grounding order), branches to the pulled-from-service claim, the maintenance-log evidence, the tribunal’s damages award and the regulator’s bulletin, and one EQUIVALENCE edge — plus two conflicts. Every value below is exact, by 2^{16} enumeration, and the graph is reproduced edge-by-edge in Figure 5 with each edge’s weight shown by line thickness.

Example 4 (Incoherent). *The report asserts both that no passengers or crew were harmed and that a wire report claimed three passengers died. These are exhaustive alternatives rather than mere opposition: they cannot both hold and cannot both fail, so the relation is EXCLUSIVE with $p = 0.93$ —*

and nothing in the response resolves it. A second tension, between the airline’s pilot-error argument and the final report’s metallurgical-defect finding ($p = 0.80$), is a **Concession** that the tribunal’s holding does resolve, and Eq. 2 discounts it accordingly. The model returns $\text{LCS}_{\text{mm}} = 0.607$ with $\log Z = -9.64$. The diagnostic that localizes the incoherence is per-claim: the marginal of the claim on the losing side of the unresolved conflict is dragged below its own prior, which no aggregate score would reveal. Applying the concession discount alone — resolving the second tension while leaving the first — raises the score to 0.612, as C2 requires.

Example 5 (Coherent). Removing the two conflict edges and changing nothing else leaves the identical support structure (Figure 5b). The score rises to $\text{LCS}_{\text{mm}} = 0.620$ with $\log Z = -8.25$ and $\text{LCS}_{\text{lp}} = 1.00$: the response is now as coherent as its own edge skeleton permits, which is what the ceiling of Eq. 7 means. The suppressed claim recovers to its prior.

Because the two panels of Figure 5 differ *only* in the two conflict edges, the $0.607 \rightarrow 0.620$ lift and the 1.39 nats of recovered probability mass are attributable to conflict structure alone — which is the property the ladder construction of §4.1 generalizes from one pair to a family. The full ladder over this family is monotone throughout: $0.586 < 0.607 < 0.612 < 0.613 < 0.620$ over the rungs *worse, base, concession-resolved, one-conflict-fixed, coherent*.¹

F.3 A CLAIM-IDENTICAL ORDERING PAIR

This pair makes the order-sensitivity claim concretely. Two summaries of one appellate judgment are built from an *identical* set of eighteen claims — the same atom texts, one a permutation of the other — differing only in whether self-serving assertions precede or follow the findings that qualify them. The faithful ordering scores $\text{LCS}_{\text{mm}} = 0.95$ and the self-serving-first ordering 0.90. No per-claim factuality check can separate them, because every claim is byte-identical; what differs is which claims the response leaves standing unqualified at the point where it asserts them.

It is worth being exact about *how* the two scores differ, because Remark 2 might seem to forbid it: nothing in Eq. 1 references position, so an edit preserving the relation set cannot move any readout. The pair does not preserve the relation set. Mining is response-grounded (§3.5), so the relations the estimator draws are a function of the prose, not of the claim list: the faithful summary states each tension beside the truth or holding that settles it, which the estimator reads as a **Concession** the text resolves and Eq. 2 discounts; the adversarial summary leaves the same claims standing unqualified until a later holding overrides them, which reads as a live conflict. The claim set is what factuality sees and it is identical; the relation set is what coherence sees and it is not.

G k -ARY REPRESENTABILITY

This appendix proves Prop. 2 of §4.6 and quantifies the gap it describes.

Proof of Prop. 2. By Eq. 1, $\mathbb{P}(1, 1, 1) = Z^{-1} \prod_i \phi_i(1) \prod_{i < j} \psi_{ij}(1, 1)$, a product of non-negative terms, which vanishes only if some term does. If $\phi_i(1) = 0$ then a_i is false almost surely and both pairs containing i have probability zero. If $\psi_{ij}(1, 1) = 0$ then every world with $a_i = a_j = 1$ carries that factor, so $\mathbb{P}(a_i=1, a_j=1) = 0$ directly. A ternary factor that vanishes only at $(1, 1, 1)$ has neither consequence. $\square$

The gap is quantitative as well as qualitative. With uniform priors, a ternary factor supported off $(1, 1, 1)$ gives $\mathbb{P}(1, 1, 1) = 0$ while leaving each pair jointly true with probability 0.143; the best pairwise attempt that drives $\mathbb{P}(1, 1, 1)$ below 10^{-4} leaves the pairs at 0.028, and a random search over 2×10^5 pairwise parameterizations found nothing better. The pairwise model can only suppress the bad world by also suppressing the good ones that share a $(1, 1)$ cell with it. Every figure here is exact by 2^3 enumeration and reproduced by a released script and regression test, so the proposition is checked and not merely asserted.

¹Treating both tensions as plain CONTRADICTION rather than EXCLUSIVE — the three-coupling model that omits the exhaustiveness distinction — gives $\text{LCS}_{\text{mm}} = 0.587$ and $\log Z = -9.75$ for Example 4, with Example 5 unchanged at 0.620 and -8.25 . The distinction of §3.2 is worth 0.020 of score here, and the reason it matters more than that figure suggests is Table 8: with EXCLUSIVE conflicts, readout (b)’s conflict term is pinned at 1.000 and the readout fails outright.

On the three-claim arithmetic instance of §4.6 — a league of 356 teams forming 32 conferences of exactly 10 members — the readouts report the passage as coherent at $\text{LCS}_{\text{mm}} = 0.68$ and $\text{LCS}_{\text{lp}} = 1.00$. Prop. 2 says no reweighting of the pairwise factors could have helped. Of the two milder consequences named in §4.6, transitivity is the more visible: the model will happily hold $a \Rightarrow b$ and $b \Rightarrow c$ at high weight while leaving a and c unrelated, since it has no rule tying them.

H CORRESPONDENCE WITH MARKOV LOGIC

The coherence MRF has a natural first-order encoding (Richardson & Domingos, 2006), with $\mathbb{P}(\mathbf{a}) = \frac{1}{Z} \exp(\sum_k w_k n_k(\mathbf{a}))$ for $n_k(\mathbf{a})$ the number of true groundings of formula F_k . Evidence predicates *Entail*(i, j), *Contradict*(i, j), *Equiv*(i, j) and *Resolves*(h, i, j) carry the mined graph; *Holds*(i) is the query predicate. Working out the correspondence is instructive in both directions.

H.1 THE NAIVE ENCODING IS A DIFFERENT MODEL

The obvious encoding gives each relation one weighted clause, e.g. $\text{Entail}(i, j) \wedge a_i \Rightarrow a_j$ with $w = \text{logit}(p)$. This is *not* equivalent to Definition 1. Its induced pairwise factor is $[e^w, e^w, 1, e^w]$, which is not row-stochastic in the sense of Prop. 1: it rewards the vacuous $a_s = 0$ rows with e^w on *both* target states, so a clause is best satisfied by making its antecedent false. Brute force over the 2^{16} worlds of the recall report of §F.2 gives mean marginal support 0.455 under the naive MLN against 0.587 under the corresponding MRF, with per-claim marginals differing by up to 0.31. These are two different models, not two encodings of one.

The failure is sharpest at the mode. The all-false world satisfies every implication vacuously, so under the naive encoding the most probable configuration of any purely-implicational graph is the one in which the response asserts nothing — the maximum-a-posteriori reading of a response is that none of it holds. This is Prop. 3 appearing in a second guise, and it is the same lesson: an encoding that scores rule satisfaction rather than belief can be maximized by having nothing to satisfy.

H.2 A THREE-CLAUSE EXPANSION IS EXACTLY EQUIVALENT

Writing the log-potential in the general pairwise form $\ln \psi(a_s, a_t) = \alpha + \beta a_s + \gamma a_t + \delta a_s a_t$ and matching it against the with-priors tables of Table 7 gives, for π_s the source prior:

τ	α (unit)	β (source)	γ (target)	δ (conjunction)
ENTAILMENT	$\ln \frac{1-p}{\pi_s}$	0	0	$+\text{logit}(p)$
CONTRADICTION	$\ln \frac{p}{\pi_s}$	0	0	$-\text{logit}(p)$
EQUIVALENCE	$\ln \frac{1-p}{p}$	$\ln \frac{1-p}{p}$	0	$+2 \text{logit}(p)$

The conjunction weight is exactly $\pm \text{logit}(p)$ — the interaction the implication was meant to carry all along — and the unit clause is the correction that restores row-stochasticity. With this expansion the ground Markov logic network reproduces the MRF exactly: mean support 0.5866 and $\log Z = -9.751$ for both, with zero per-claim difference to machine precision.

We establish the equivalence for the three original couplings only. EXCLUSIVE and CO_NECESSITY are not tabulated here, so the claim “the MLN encoding reproduces the MRF” should be read as scoped to the three-coupling fragment. That is a real limitation for the recall report of §F.2, whose conflicts are EXCLUSIVE; the five-claim pair of §F.1 uses CONTRADICTION throughout and so falls inside the tabulated fragment.

H.3 WHICH VIEW TO KEEP, AND WHY BOTH

The reason to prefer the MRF for the measure itself is Remark 1: its pairwise factor family is closed at three parameters, so the inferential vocabulary cannot quietly grow as senses are added, and every relation type is guaranteed to be a 2×2 potential. The reason to keep the Markov logic view is that the rules our model cannot express are naturally written there — a transitivity rule $\text{Entail}(i, j) \wedge \text{Entail}(j, k) \Rightarrow \text{Entail}(i, k)$, a concession-cancellation rule $\text{Resolves}(h, i, j) \wedge a_h \Rightarrow \neg \text{Penalize}(i, j)$ replacing the fixed discount of Eq. 2, and a double-conflict rule — and their weights

could be *learned* from labelled ladders rather than set by hand. That is the route past the two limitations §4.6 names, and it requires an MLN solver (MC-SAT for marginals, MaxWalkSAT for the mode) rather than the bucket-elimination stack used here.

I THE REIFIED COHERENCE NODE, WORKED

To make Eq. 6 concrete, take the three-claim subnet consisting of the unresolved casualty conflict of Example 4 together with one entailment feeding it: claims a_7, a_{10} joined by a conflict at $p = 0.93$, and $a_4 \rightarrow a_7$ an entailment at $p = 0.65$, all priors $\frac{1}{2}$, with $\rho = \frac{1}{2}$. Writing out the eight-cell vote factor for each relation from Eq. 6 and enumerating the 2^4 worlds of (a_4, a_7, a_{10}, R) gives $\mathbb{P}(R=1) = 0.453$: below its prior, because both relations have violating worlds that carry appreciable mass, and the conflict's violating world $(1, 1)$ carries the most.

Two properties of the construction are worth noting. The $R = 0$ branch is flat, so $R = 0$ asserts nothing and the node cannot be driven up merely by adding relations that happen to be satisfied. And the factor is a *noisy* AND rather than a hard one: a relation with $p_r < 1$ leaves some mass on its violating worlds even when $R = 1$, so a single strongly-violated relation reduces $\mathbb{P}(R=1)$ without pinning it to zero. The reason we do not adopt the readout despite this reasonable behaviour is Table 8: it dips at the concession variant for the reason of Prop. 4, and it requires the free parameter ρ , violating R4.

J THE ESTIMATION PROMPTS, VERBATIM

The four prompts below are reproduced exactly as they are issued, extracted programmatically from the released source so that this appendix cannot drift from the code. Placeholders in double braces are substituted at call time; `{{sense_menu}}` is substituted at build time from one of the two menus in §J.5.

J.1 PROMPT A, FIRST GENERATION — DISCOURSE SENSE AND COUPLING

6,148 characters. Placeholders: `{{sense_menu}}`, `{{response}}`, `{{atom_a}}`, `{{atom_b}}`.

```

1
2
3 Instructions:
4 You are given the full RESPONSE a model produced, and two atomic claims, A and B, both
  ↳ taken FROM THAT RESPONSE, in their order of appearance (A comes before B). Your task is
  ↳ to decide the discourse/logical relation FROM A TO B, following the steps below.
5
6 IMPORTANT -- ground your decision in the response. Assert a coupling ONLY if the response
  ↳ ITSELF draws that connection between A and B (as written, or as a clear step in the
  ↳ author's argument/narrative). Do NOT assert a relation that is merely plausible in
  ↳ general but that the response does not actually make. If A and B both appear in the
  ↳ response yet the response draws no logical or discourse dependence between them, the
  ↳ answer is None.
7
8 1. Reason step by step, referring to the response: does the response present A as causing,
  ↳ enabling, providing evidence for, restating, elaborating, temporally preceding,
  ↳ contrasting with, or contradicting B? Is B a claim the response later withdraws or that
  ↳ a holding resolves? Consider the direction (A to B). If the response links A and B only
  ↳ indirectly through other claims, or not at all, that is None.
9
10 2. Name the DISCOURSE SENSE, one of: {{sense_menu}}.
11   - Cause-Effect: A causes/leads to B. Effect-Cause: A is the effect, B its cause.
12   - Evidence: A provides evidence for B. Condition: A is a condition for B.
13   - Restatement: A and B assert the same thing. Instantiation: A is a general claim, B a
  ↳ specific instance (or vice versa).
14   - Contrast: A and B are in opposition but NOT exhaustive (they need not cover all
  ↳ possibilities; both could conceivably be false).
15   - Concession: A and B are in tension but the text concedes/resolves it ("although A,
  ↳ still B", or a holding settles it).
16   - Alternative: A and B are EXHAUSTIVE competing options -- EXACTLY ONE holds (they are
  ↳ mutually exclusive AND together cover the possibilities: not both, and not neither).
  ↳ E.g. "no one was harmed" vs "three people died"; "the cause was pilot error" vs "the
  ↳ cause was a metallurgical defect".

```

1728 17 - Disjunction: AT LEAST ONE of A and B holds (they may both hold, but the response rules
1729 ↳ out neither being true) -- e.g. two supporting findings at least one of which must
1730 ↳ be present.

1731 18 - Precedence/Succession: A and B are ordered in time with no truth dependence.
1732 19 - None: the response draws no logical or discourse dependence between A and B.

1733 20

1734 21 3. Map the sense to a COUPLING, one of: entailment, contradiction, equivalence, exclusive,
1735 ↳ co_necessity, none.
1736 22 - Cause-Effect, Effect-Cause, Evidence, Condition, Instantiation -> entailment
1737 23 - Restatement -> equivalence
1738 24 - Contrast, Concession -> contradiction
1739 25 - Alternative -> exclusive (exactly one of A, B is true)
1740 26 - Disjunction -> co_necessity (at least one of A, B is true)
1741 27 - Precedence, Succession, None -> none
1742 28 Prefer "exclusive" over "contradiction" when the two claims are not just incompatible
1743 ↳ but EXHAUSTIVE (one of them must be true); prefer "contradiction" when they merely
1744 ↳ cannot both hold but could both be false.

1745 29

1746 30 4. Give your final answer as two bracketed tags on ONE line, sense first:
1747 31 [sense=Cause-Effect] [coupling=entailment]
1748 32 A JSON object {"sense":"Cause-Effect","coupling":"entailment"} is also acceptable.

1749 33

1750 34 Use the following examples to better understand your task.

1751 35

1752 36 Example 1 (the response makes the causal link):
1753 37 RESPONSE: The company launched a flawed product last quarter. Reviewers panned it, returns
1754 ↳ spiked, and the company's stock price fell 15 percent over the same period.
1755 38 A: The company launched a flawed product last quarter.
1756 39 B: The company's stock price fell 15 percent last quarter.
1757 40 1. Reasoning: the response presents the flawed launch as the head of a chain (panning,
1758 ↳ returns) that ends in the stock decline, so the response itself draws a causal link
1759 ↳ from A to B.
1760 41 2. Discourse sense: Cause-Effect.
1761 42 3. Coupling: A causing B is a positive inferential link, i.e. entailment.
1762 43 4. Final answer:
1763 44 [sense=Cause-Effect] [coupling=entailment]

1764 45

1765 46 Example 2 (both claims present, but the response draws NO connection -> None):
1766 47 RESPONSE: The quarterly report was published in April. Separately, the annual audit was
1767 ↳ scheduled for December. The two processes are run by different teams and were not
1768 ↳ related this year.
1769 48 A: The quarterly report was published in April.
1770 49 B: The annual audit was scheduled for December.
1771 50 1. Reasoning: both claims appear in the response, and one might imagine a
1772 ↳ reporting-to-audit link in general, but this response explicitly treats them as
1773 ↳ separate and unrelated. The response draws no dependence from A to B.
1774 51 2. Discourse sense: None.
1775 52 3. Coupling: no dependence the response asserts, i.e. none.
1776 53 4. Final answer:
1777 54 [sense=None] [coupling=none]

1778 55

1779 56 Example 3 (the response states an EXHAUSTIVE alternative -> exclusive):
1780 57 RESPONSE: The official statement said no one was harmed in the incident. However, the
1781 ↳ coroner's report confirmed that three people died in the incident.
1782 58 A: No one was harmed in the incident.
1783 59 B: Three people died in the incident.
1784 60 1. Reasoning: the response sets A and B against each other ("However, ...") and they cannot
1785 ↳ both be true; but they also cannot both be false -- either people were harmed or they
1786 ↳ were not -- so exactly one holds. This is exhaustive, not a mere contrast. No holding
1787 ↳ resolves it.
1788 61 2. Discourse sense: Alternative.
1789 62 3. Coupling: exactly one of A, B is true, i.e. exclusive.
1790 63 4. Final answer:
1791 64 [sense=Alternative] [coupling=exclusive]

1792 65

1793 66 Example 4 (at least one must hold -> co_necessity):
1794 67 RESPONSE: The defect was caught in review: at least one of the two independent checks --
1795 ↳ the vibration analysis or the metallurgical assay -- flagged it.
1796 68 A: The vibration analysis flagged the defect.
1797 69 B: The metallurgical assay flagged the defect.
1798 70 1. Reasoning: the response asserts the defect WAS caught by at least one check, so A and B
1799 ↳ cannot both be false; but both could hold (both checks may have flagged it). This is a
1800 ↳ disjunction, not an exclusion.
1801 71 2. Discourse sense: Disjunction.
1802 72 3. Coupling: at least one of A, B is true, i.e. co_necessity.
1803 73 4. Final answer:
1804 74 [sense=Disjunction] [coupling=co_necessity]

1805 75

1806 76 Your task:
1807 77 RESPONSE: {{response}}

78 A: {{atom_a}}
79 B: {{atom_b}}

J.2 PROMPT A, SECOND GENERATION — REBALANCED FOR PRECISION

6,361 characters. Same placeholders. The five changes relative to the first generation, and their motivation, are given in §3.5: few-shot balance from 3:1 related to 2:3; negative exemplars showing silence rather than explicit disavowal; a verifiable evidence span; the discriminating question asked first; and NONE given first position and equal prose. Transitivity is named as a non-relation.

```

1
2
3 Instructions:
4 You are given the full RESPONSE a model produced, and two atomic claims, A and B, both
5 ↳ taken FROM THAT RESPONSE. Decide the discourse/logical relation FROM A TO B.
6
7 Most pairs of claims in a text are NOT related. They are simply near each other. Your
8 ↳ default answer is None, and you should depart from it only when the response itself
9 ↳ draws a dependence between A and B.
10
11 Three things that are NOT relations, however plausible they look:
12   - Topical adjacency. A and B being about the same subject, or sitting in neighbouring
13     ↳ sentences, is not a relation.
14   - World knowledge. If the link is true in general but the response does not draw it, the
15     ↳ answer is None.
16   - Transitivity. If the response links A to some third claim C, and C to B, that does NOT
17     ↳ make A related to B. Report only the links the text draws directly.
18
19 1. First ask the discriminating question: does the response ITSELF assert a dependence from
20   ↳ A to B -- as written, or as an explicit step in the author's argument? Identify the
21   ↳ words in the response that do it. If you cannot point to such words, the answer is None.
22
23 2. If and only if you can, name the DISCOURSE SENSE, one of: {{sense_menu}}.
24   - None: the response draws no dependence between A and B. This is the common case:
25     ↳ unrelated, merely adjacent, related only by world knowledge, or related only through
26     ↳ a third claim.
27   - Cause-Effect: the response says A causes/leads to B. Effect-Cause: A is the effect, B
28     ↳ its cause.
29   - Evidence: the response offers A as evidence/support for B.
30   - Restatement: A and B say the same thing -- a paraphrase, a converse, or a restatement
31     ↳ the text marks ("in other words", "equivalently", "that is").
32   - Contrast: A and B are opposed but NOT exhaustive (both could be false).
33   - Concession: A and B are in tension and the text concedes or resolves it ("although A,
34     ↳ still B"; a holding settles it).
35   - Alternative: A and B are EXHAUSTIVE competing options -- exactly one holds (not both,
36     ↳ not neither).
37   - Disjunction: AT LEAST ONE of A and B holds; both may.
38   - Precedence: A and B are ordered in time with no truth dependence.
39
40 3. Map the sense to a COUPLING, one of: entailment, contradiction, equivalence, exclusive,
41   ↳ co_necessity, none.
42   - Cause-Effect, Effect-Cause, Evidence -> entailment
43   - Restatement -> equivalence
44   - Contrast, Concession -> contradiction
45   - Alternative -> exclusive (exactly one of A, B is true)
46   - Disjunction -> co_necessity (at least one of A, B is true)
47   - Precedence, None -> none
48   Prefer "exclusive" when the two claims are incompatible AND exhaustive; "contradiction"
49   ↳ when they merely cannot both hold but could both be false.
50
51 4. Answer on ONE line, with three bracketed tags. The evidence must be a SHORT VERBATIM
52   ↳ QUOTE COPIED FROM THE RESPONSE -- the words that draw the link. Do not quote claim A or
53   ↳ claim B; quote the response. Use [evidence=none] when the sense is None.
54   [sense=Evidence] [coupling=entailment] [evidence="which is why the panel held"]
55
56 Use the following examples to better understand your task.
57
58 Example 1 (adjacent and on-topic, but the response draws no link -> None):
59 RESPONSE: The survey was administered in March to a sample of 400 households. Response
60 ↳ rates in rural districts have declined over the past decade. The analysis weighted
61 ↳ results by district population.
62 A: The survey was administered in March to a sample of 400 households.
63 B: Response rates in rural districts have declined over the past decade.
64 1. The two sentences are adjacent and both concern the survey, but the response never says
65 ↳ the March administration bears on rural response rates. There are no words that draw a
66 ↳ dependence.
67 2. Discourse sense: None.

```

```

1836 46 3. Coupling: none.
1837 47 4. [sense=None] [coupling=none] [evidence=none]
1838 48
1839 49 Example 2 (the response makes the causal link explicitly):
1840 50 RESPONSE: The company launched a flawed product last quarter. Reviewers panned it, returns
1841 51 ↳ spiked, and as a direct result the company's stock price fell 15 percent over the same
1842 52 ↳ period.
1843 53 A: The company launched a flawed product last quarter.
1844 54 B: The company's stock price fell 15 percent last quarter.
1845 55 1. The response presents the launch as the head of a chain ending in the stock decline, and
1846 56 ↳ marks it: "as a direct result".
1847 57 2. Discourse sense: Cause-Effect.
1848 58 3. Coupling: entailment.
1849 59 4. [sense=Cause-Effect] [coupling=entailment] [evidence="as a direct result"]
1850 60
1851 61 Example 3 (true in general, but this response does not draw it -> None):
1852 62 RESPONSE: Higher interest rates raise the cost of corporate borrowing. The firm opened
1853 63 ↳ three distribution centres last year. Its logistics costs fell by eight percent.
1854 64 A: Higher interest rates raise the cost of corporate borrowing.
1855 65 B: The firm opened three distribution centres last year.
1856 66 1. One could argue borrowing costs affect expansion decisions, but that is world knowledge.
1857 67 ↳ This response states the two facts side by side and links neither to the other.
1858 68 2. Discourse sense: None.
1859 69 3. Coupling: none.
1860 70 4. [sense=None] [coupling=none] [evidence=none]
1861 71
1862 72 Example 4 (the response states an exhaustive alternative -> exclusive):
1863 73 RESPONSE: The two readings are mutually exclusive and exactly one must be right: either no
1864 74 ↳ one was harmed in the incident, or three people died in it.
1865 75 A: No one was harmed in the incident.
1866 76 B: Three people died in the incident.
1867 77 1. The response sets A and B against each other and says exactly one must be right:
1868 78 ↳ "mutually exclusive and exactly one must be right: either ... or".
1869 79 2. Discourse sense: Alternative.
1870 80 3. Coupling: exclusive.
1871 81 4. [sense=Alternative] [coupling=exclusive] [evidence="mutually exclusive and exactly one
1872 82 ↳ must be right"]
1873 83
1874 84 Example 5 (linked only through a third claim -> None):
1875 85 RESPONSE: The alloy is chemically identical to the certified reference. It therefore meets
1876 86 ↳ the reference specification. Meeting that specification is a precondition for airframe
1877 87 ↳ use.
1878 88 A: The alloy is chemically identical to the certified reference.
1879 89 B: Meeting that specification is a precondition for airframe use.
1880 90 1. The response links A to the middle claim (it meets the specification), and the middle
1881 91 ↳ claim to B. It does not link A to B directly; that would be transitivity, which is not
1882 92 ↳ a relation to report.
1883 93 2. Discourse sense: None.
1884 94 3. Coupling: none.
1885 95 4. [sense=None] [coupling=none] [evidence=none]
1886 96
1887 97 Your task:
1888 98 RESPONSE: {{response}}
1889 99 A: {{atom_a}}
1900 100 B: {{atom_b}}

```

J.3 PROMPT B, DEFAULT — SURROGATE-TOKEN CONDITIONAL STRENGTH

2,182 characters. Placeholders: {{response}}, {{atom_a}}, {{atom_b}}, {{coupling}} (twice). This is the one prompt that does not end in a newline: it terminates with Answer: including the trailing space, so that the model's next token is the surrogate whose logprobs are read. Reproducing it without that trailing space changes the tokenization of the answer position and therefore the estimate.

```

1882 1
1883 2
1884 3 Instructions:
1885 4 You are given the full RESPONSE, two claims A and B drawn from it, and a {{coupling}}
1886 5 ↳ relation that holds from A to B. Assuming A is TRUE, judge -- IN THE CONTEXT OF THIS
1887 6 ↳ RESPONSE -- whether the relation's implication about B is credible: at least plausible
1888 7 ↳ / more likely than not, NOT whether it is certain.
1889 8
1890 9 - entailment or equivalence: given A and how the response uses it, is B at least plausibly
1891 10 ↳ TRUE (more likely than not)?
1892 11 - contradiction or exclusive: given A and how the response uses it, is B at least plausibly
1893 12 ↳ FALSE (more likely than not)? (For "exclusive", A and B are exhaustive alternatives, so
1894 13 ↳ A being true makes B false.)

```

```

1890 8 - co_necessity: A and B are a pair of which at least one holds. Given the response rules
1891 9 ↪ out "neither", is it at least plausible that B holds when A does NOT?
1892 10
1893 11 Answer with a SINGLE WORD, the very first word of your reply: Yes or No.
1894 12 - Answer "Yes" if the implication is credible/plausible (even if not certain).
1895 13 - Answer "No" only if the implication is implausible or the response does not actually
1896 14 ↪ support it.
1897 15 Do not output anything before the word Yes or No.
1898 16
1899 17 Example 1 (response supports a near-certain entailment):
1900 18 RESPONSE: The new alloy is chemically identical to the certified reference alloy, so it
1901 19 ↪ meets the certified reference specification.
1902 20 A: The new alloy is chemically identical to the certified reference alloy.
1903 21 B: The new alloy meets the certified reference specification.
1904 22 coupling: entailment
1905 23 Answer: Yes
1906 24
1907 25 Example 2 (weak but plausible, and the response draws the link -- still Yes):
1908 26 RESPONSE: The company launched a flawed product last quarter, and its stock price fell 15
1909 27 ↪ percent over the same period.
1910 28 A: The company launched a flawed product last quarter.
1911 29 B: The company's stock price fell 15 percent last quarter.
1912 30 coupling: entailment
1913 31 Answer: Yes
1914 32
1915 33 Example 3 (clear contradiction stated by the response):
1916 34 RESPONSE: The official statement said no one was harmed, but three people died in the
1917 35 ↪ incident.
1918 36 A: No one was harmed in the incident.
1919 37 B: Three people died in the incident.
1920 38 coupling: contradiction
1921 39 Answer: Yes
1922 40
1923 41 Your task:
1924 42 RESPONSE: {{response}}
1925 43 A: {{atom_a}}
1926 44 B: {{atom_b}}
1927 45 coupling: {{coupling}}
1928 46 Answer:

```

J.4 PROMPT B, BASELINE — VERBALIZED CONDITIONAL STRENGTH

2,112 characters. Retained only for the comparison reported in §4; verbalized confidence is weakly calibrated (Xiong et al., 2024).

```

1923 1
1924 2
1925 3 Instructions:
1926 4 You are given the full RESPONSE and two claims A and B drawn from it. Assume claim A is
1927 5 ↪ TRUE and, IN THE CONTEXT OF THIS RESPONSE, estimate how strongly A determines B under a
1928 6 ↪ {{coupling}} relation from A to B.
1929 7
1930 8 - For an entailment or equivalence coupling: how likely is B to be TRUE given A?
1931 9 - For a contradiction or exclusive coupling: how likely is B to be FALSE given A?
1932 10 ↪ (exclusive = A and B are exhaustive alternatives, so A true forces B false.)
1933 11 - For a co_necessity coupling (at least one of A, B holds): how likely is B to be TRUE when
1934 12 ↪ A is FALSE?
1935 13
1936 14 Answer with a single probability in [0, 1] to two decimals, after one short sentence of
1937 15 ↪ justification. A near-certain link is close to 1.00; a merely plausible link is around
1938 16 ↪ 0.60-0.70. End your answer with the probability wrapped in brackets on its own, exactly
1939 17 ↪ in the form: [p=0.NN]
1940 18
1941 19 Example 1:
1942 20 RESPONSE: The new alloy is chemically identical to the certified reference alloy, so it
1943 21 ↪ meets the certified reference specification.
1944 22 A: The new alloy is chemically identical to the certified reference alloy.
1945 23 B: The new alloy meets the certified reference specification.
1946 24 coupling: entailment
1947 25 Justification: chemical identity to a certified reference almost guarantees the
1948 26 ↪ specification is met, so B follows very strongly from A.
1949 27 [p=0.95]
1950 28
1951 29 Example 2:
1952 30 RESPONSE: The company launched a flawed product last quarter, and its stock price fell 15
1953 31 ↪ percent over the same period.
1954 32 A: The company launched a flawed product last quarter.

```

1944 23 B: The company's stock price fell 15 percent last quarter.
 1945 24 coupling: entailment
 1946 25 Justification: a flawed product can plausibly drive a stock decline, but many other factors
 1947 26 $\hookrightarrow$ affect price, so the link is only moderate.
 1948 27 [p=0.65]
 1949 28 Example 3:
 1949 29 RESPONSE: The official statement said no one was harmed, but three people died in the
 1950 30 $\hookrightarrow$ incident.
 1951 31 A: No one was harmed in the incident.
 1951 32 B: Three people died in the incident.
 1952 33 coupling: contradiction
 1952 34 Justification: if no one was harmed, it is almost certain that B (three deaths) is false.
 1953 35 [p=0.93]
 1954 36 Your task:
 1955 37 RESPONSE: {{response}}
 1956 38 A: {{atom_a}}
 1957 39 B: {{atom_b}}
 1957 40 coupling: {{coupling}}

1958 J.5 THE TWO SENSE MENUS

1959 The full menu offers all twelve senses of Table 5 plus NONE:
 1960

1961 1 Cause-Effect, Effect-Cause, Evidence, Condition, Restatement, Instantiation, Contrast,
 1962 $\hookrightarrow$ Concession, Alternative, Disjunction, Precedence, Succession, None

1963 The restricted menu offers only the senses the evaluation corpus labels, which is the admissibility
 1964 filter of §3.5 applied at elicitation time rather than after it:
 1965

1966 1 Cause-Effect, Effect-Cause, Evidence, Restatement, Contrast, Concession, Alternative,
 1967 $\hookrightarrow$ Disjunction, Precedence, None

1968 Note that the restricted menu omits **Succession** as well as **Condition** and **Instantiation**, leaving
 1969 nine relation-bearing senses. **Succession** is dropped because it is the mirror of **Precedence** and
 1970 both compile to **NONE**, so offering both invites an arbitrary choice on a relation that produces no
 1971 factor either way. The second-generation Prompt A's body is written against this restricted menu;
 1972 when it is run with the full menu, three senses are offered with no gloss and no coupling mapping in
 1973 the prompt body, and we report which menu each arm used.
 1974

1975 K SERIALIZATION AND THE VARIABLE-ORDERING TRAP

1976 One implementation detail is worth recording because it is silent when wrong. Networks are serialized
 1977 in UAI format, whose variables are identified by integer index; the index order is the sort order of
 1978 (cardinality, name), and with uniformly binary claim variables that reduces to *lexicographic order*
 1979 *by name*. So a_{10} sorts before a_2 . Inference returns marginals by index, and mapping them back by
 1980 any other convention — source order, numeric order — silently permutes marginals across claims.
 1981 The score is unchanged, since Eq. 4 is a mean, but every per-claim diagnostic is then attributed to
 1982 the wrong claim, and the “claim dragged below its prior” readout points at an innocent claim. The
 1983 auxiliary variables of §C.4 are named with a prefix that sorts after all claim names for the same
 1984 reason, so that adding them cannot shift the claim indices.
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